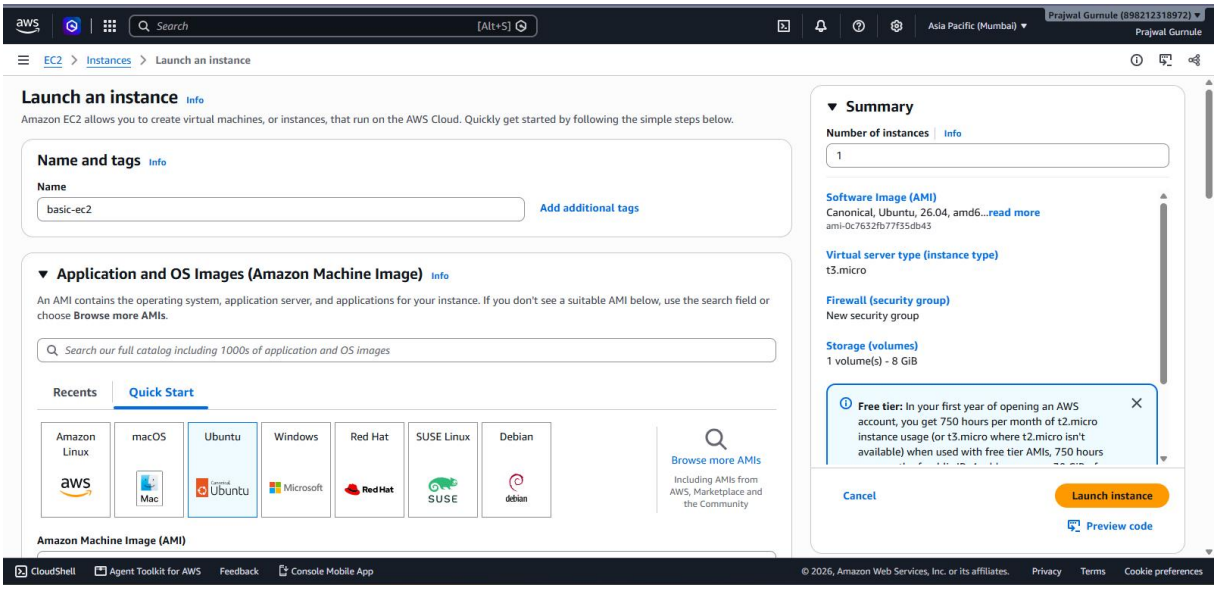
Assignment 8

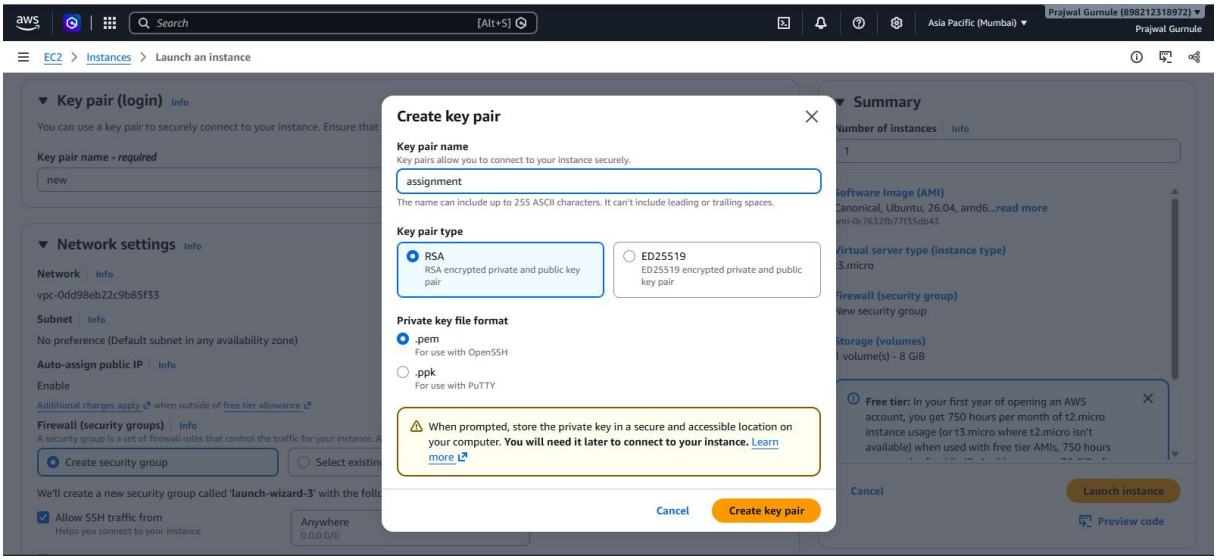
Q1. Create EC2 Instance and Connect

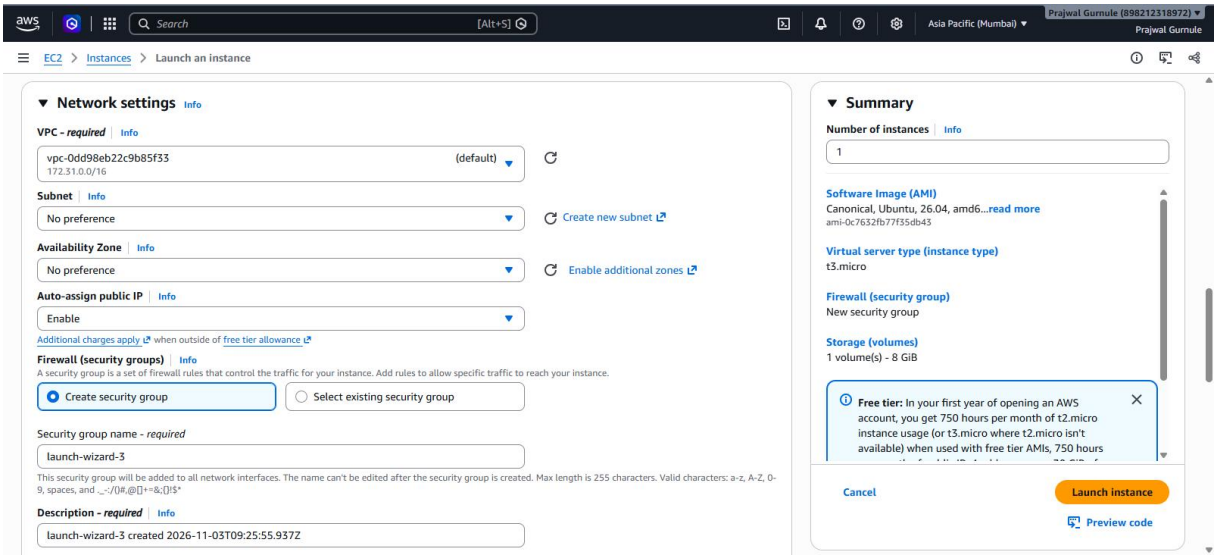
Create a basic virtual machine.

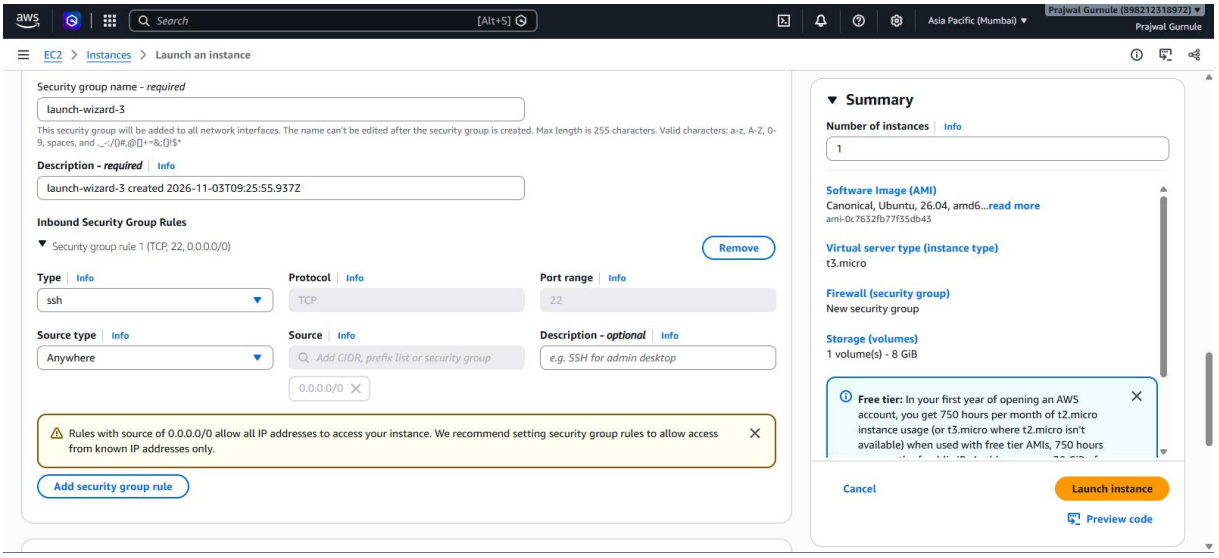
Tasks:

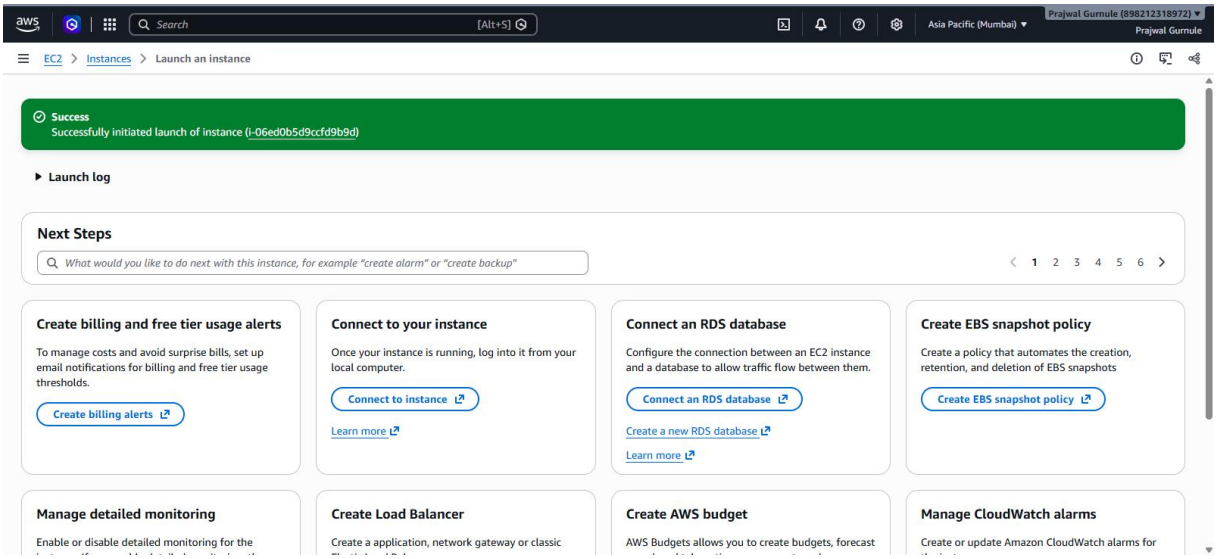
- Go to **Amazon EC2** Dashboard
- Launch instance:
  - Name: `basic-ec2`
  - AMI: Ubuntu
  - Type: `t3.micro`
- Create a new key pair and download it
- Create Security Group:
  - Allow SSH (port 22) from your IP
- Launch instance
- Connect using EC2 Instance Connect (browser)

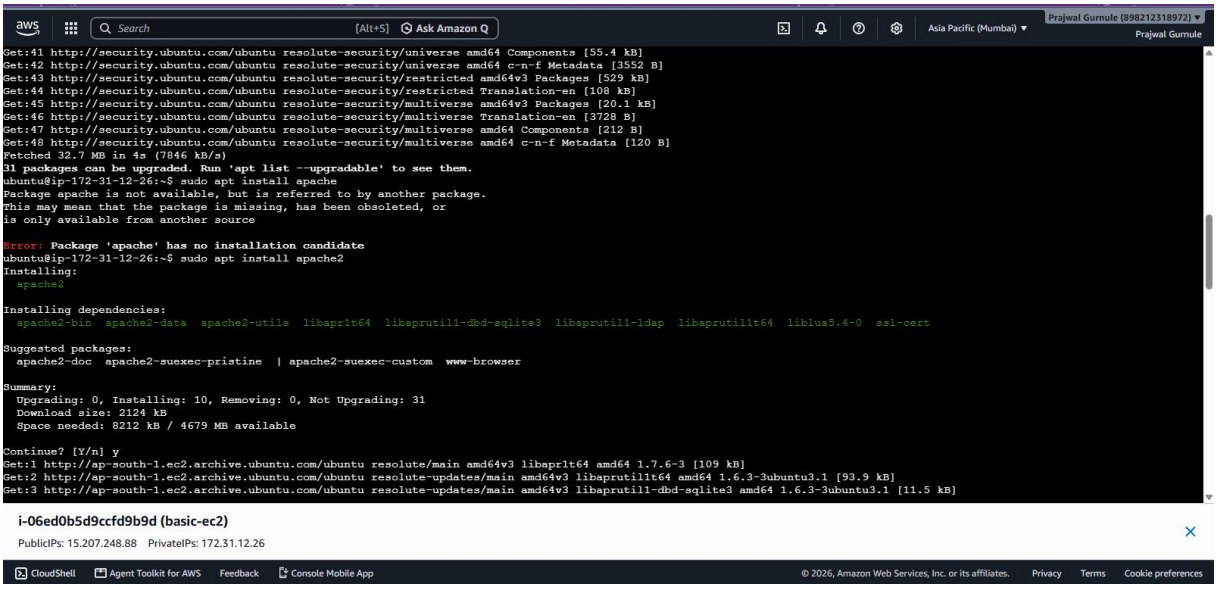


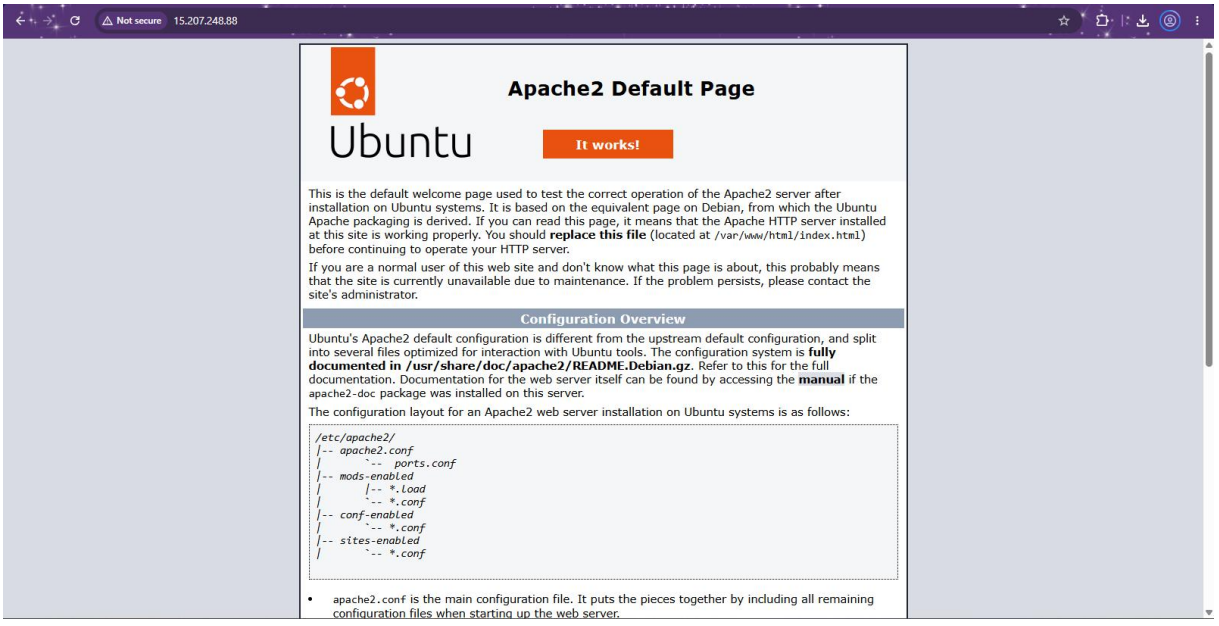

Search [Alt+S]
Asia Pacific (Mumbai)
Prajwal Gurnule (898212318972)
Prajwal Gurnule
EC2 > Instances > Launch an instance
Key pair (login) Info
You can use a key pair to securely connect to your instance. Ensure that
Key pair name - required
new
Network settings Info
Network Info
vpc-0dd98eb22c9b85f33
Subnet Info
No preference (Default subnet in any availability zone)
Auto-assign public IP Info
Enable
Additional charges apply when outside of free tier allowance
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance.
Create security group
Select existing
We'll create a new security group called 'launch-wizard-3' with the follo
Allow SSH traffic from
Helps you connect to your instance
Anywhere
0.0.0.0/0
Create key pair
Key pair name
Key pairs allow you to connect to your instance securely.
assignment
The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.
Key pair type
RSA
RSA encrypted private and public key pair
ED25519
ED25519 encrypted private and public key pair
Private key file format
.pem
For use with OpenSSH
.ppk
For use with PuTTY
When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. Learn more
Cancel
Create key pair
Summary
Number of instances Info
1
Software Image (AMI)
Canonical, Ubuntu, 26.04, amd6...read more
ami-0c7632fb77f35db43
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours
Cancel
Launch instance
Preview code


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EC2 > Instances > Launch an instance
Network settings Info
VPC - required Info
vpc-0dd98eb22c9b85f33 (default)
172.31.0.0/16
Subnet Info
No preference
Create new subnet
Availability Zone Info
No preference
Enable additional zones
Auto-assign public IP Info
Enable
Additional charges apply when outside of free tier allowance
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
Security group name - required
launch-wizard-3
This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and ._-:/()#,@[]+=&;{}!$*
Description - required Info
launch-wizard-3 created 2026-11-03T09:25:55.937Z
Summary
Number of instances Info
1
Software Image (AMI)
Canonical, Ubuntu, 26.04, amd6...read more
ami-0c7632fb77f35db43
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours
Cancel
Launch instance
Preview code


Search [Alt+S]
Asia Pacific (Mumbai)
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EC2 > Instances > Launch an instance
Security group name - required
launch-wizard-3
This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and ._-:/()#,@[]+=&;{}!$*
Description - required Info
launch-wizard-3 created 2026-11-03T09:25:55.937Z
Inbound Security Group Rules
Security group rule 1 (TCP, 22, 0.0.0.0/0)
Remove
Type Info
ssh
Protocol Info
TCP
Port range Info
22
Source type Info
Anywhere
Source Info
Add CIDR, prefix list or security group
0.0.0.0/0
Description - optional Info
e.g. SSH for admin desktop
Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.
Add security group rule
Summary
Number of instances Info
1
Software Image (AMI)
Canonical, Ubuntu, 26.04, amd6...read more
ami-0c7632fb77f35db43
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours
Cancel
Launch instance
Preview code


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Prajwal Gurnule
EC2 > Instances > Launch an instance
Success
Successfully initiated launch of instance (i-06ed0b5d9ccfd9b9d)
Launch log
Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup"
1 2 3 4 5 6
Create billing and free tier usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing and free tier usage thresholds.
Create billing alerts
Connect to your instance
Once your instance is running, log into it from your local computer.
Connect to instance
Learn more
Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
Connect an RDS database
Create a new RDS database
Learn more
Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots
Create EBS snapshot policy
Manage detailed monitoring
Enable or disable detailed monitoring for the
Create Load Balancer
Create a application, network gateway or classic
Create AWS budget
AWS Budgets allows you to create budgets, forecast
Manage CloudWatch alarms
Create or update Amazon CloudWatch alarms for


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Asia Pacific (Mumbai)
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Get:41 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [55.4 kB]
Get:42 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 c-n-f Metadata [3552 B]
Get:43 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64v3 Packages [529 kB]
Get:44 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [108 kB]
Get:45 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64v3 Packages [20.1 kB]
Get:46 http://security.ubuntu.com/ubuntu resolute-security/multiverse Translation-en [3728 B]
Get:47 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Components [212 B]
Get:48 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 c-n-f Metadata [120 B]
Fetched 32.7 MB in 4s (7846 kB/s)
31 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-172-31-12-26:~$ sudo apt install apache
Package apache is not available, but is referred to by another package.
This may mean that the package is missing, has been obsoleted, or
is only available from another source
Error: Package 'apache' has no installation candidate
ubuntu@ip-172-31-12-26:~$ sudo apt install apache2
Installing:
apache2
Installing dependencies:
apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64 liblua5.4-0 ssl-cert
Suggested packages:
apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser
Summary:
Upgrading: 0, Installing: 10, Removing: 0, Not Upgrading: 31
Download size: 2124 kB
Space needed: 8212 kB / 4679 MB available
Continue? [Y/n] y
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 libapr1t64 amd64 1.7.6-3 [109 kB]
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 libaprutil1t64 amd64 1.6.3-3ubuntu3.1 [93.9 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 libaprutil1-dbd-sqlite3 amd64 1.6.3-3ubuntu3.1 [11.5 kB]
i-06ed0b5d9ccfd9b9d (basic-ec2)
PublicIPs: 15.207.248.88 PrivateIPs: 172.31.12.26
CloudShell Agent Toolkit for AWS Feedback Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

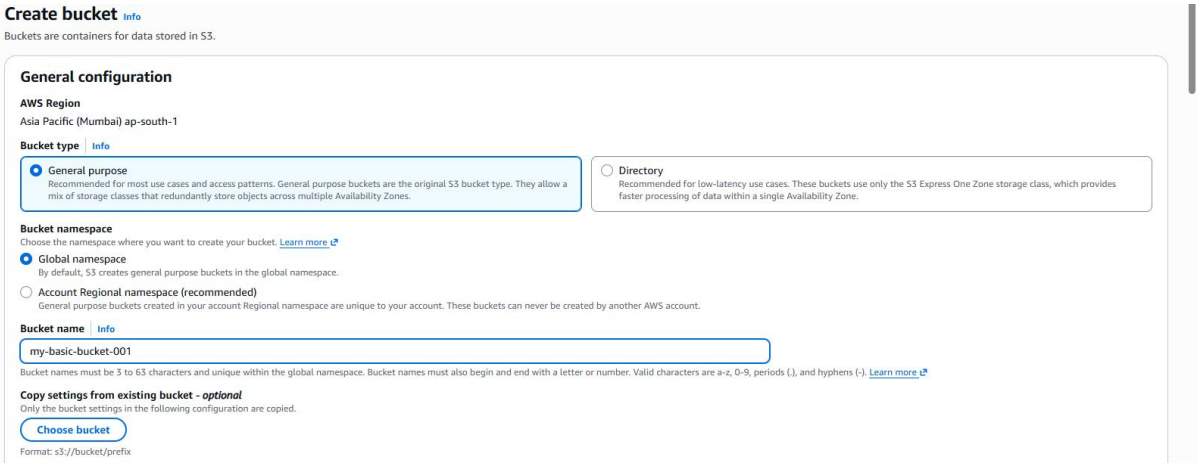

Not secure 15.207.248.88
Apache2 Default Page
Ubuntu
It works!
This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should replace this file (located at /var/www/html/index.html) before continuing to operate your HTTP server.
If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.
Configuration Overview
Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is fully documented in /usr/share/doc/apache2/README.Debian.gz. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the manual if the apache2-doc package was installed on this server.
The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:
/etc/apache2/
|-- apache2.conf
| `-- ports.conf
|-- mods-enabled
| |-- *.load
| `-- *.conf
|-- conf-enabled
| `-- *.conf
|-- sites-enabled
| `-- *.conf
apache2.conf is the main configuration file. It puts the pieces together by including all remaining configuration files when starting up the web server.

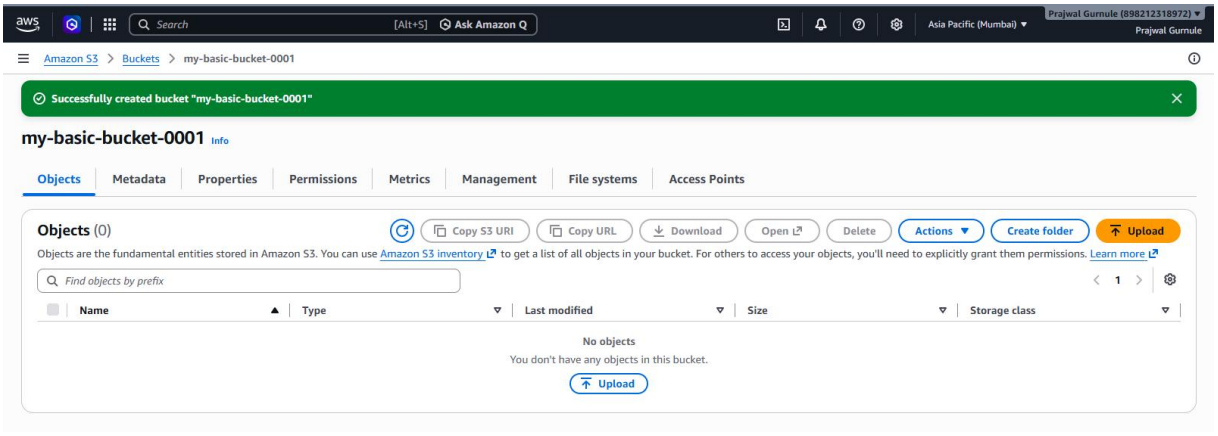
Q2. Create S3 Bucket and Upload File

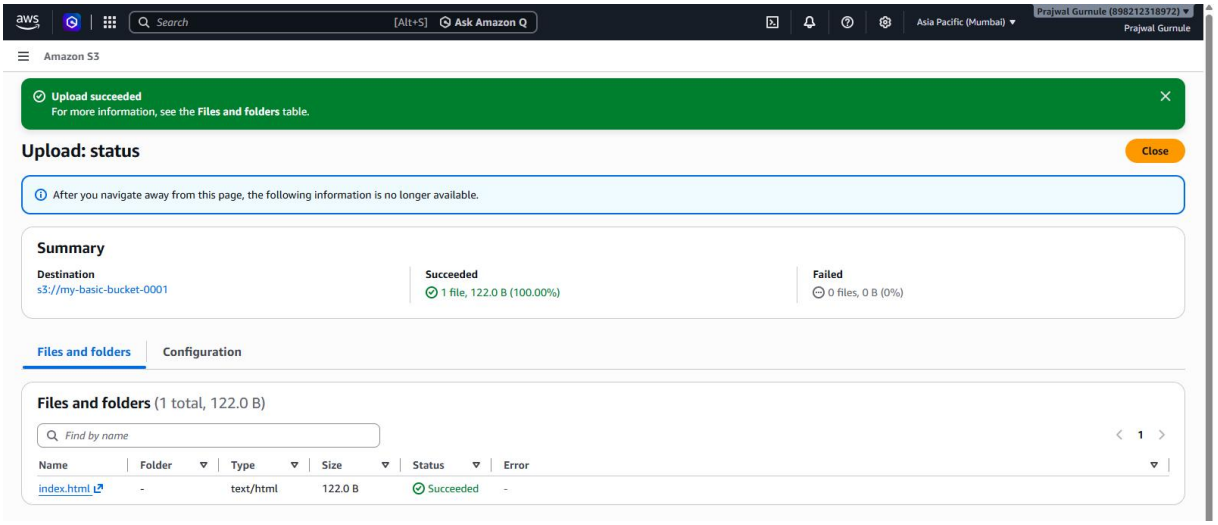
Work with storage using GUI.

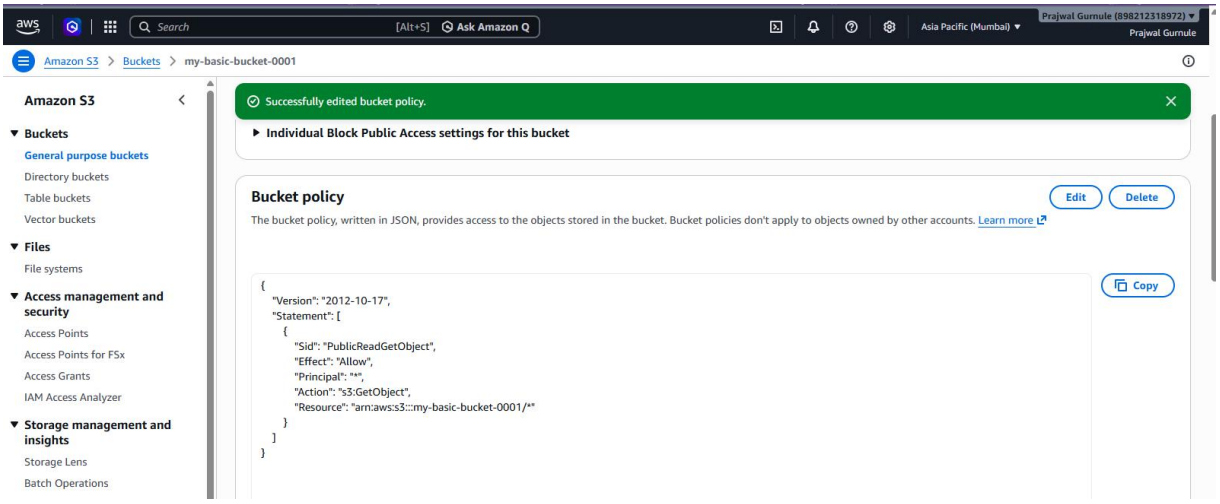
Tasks:

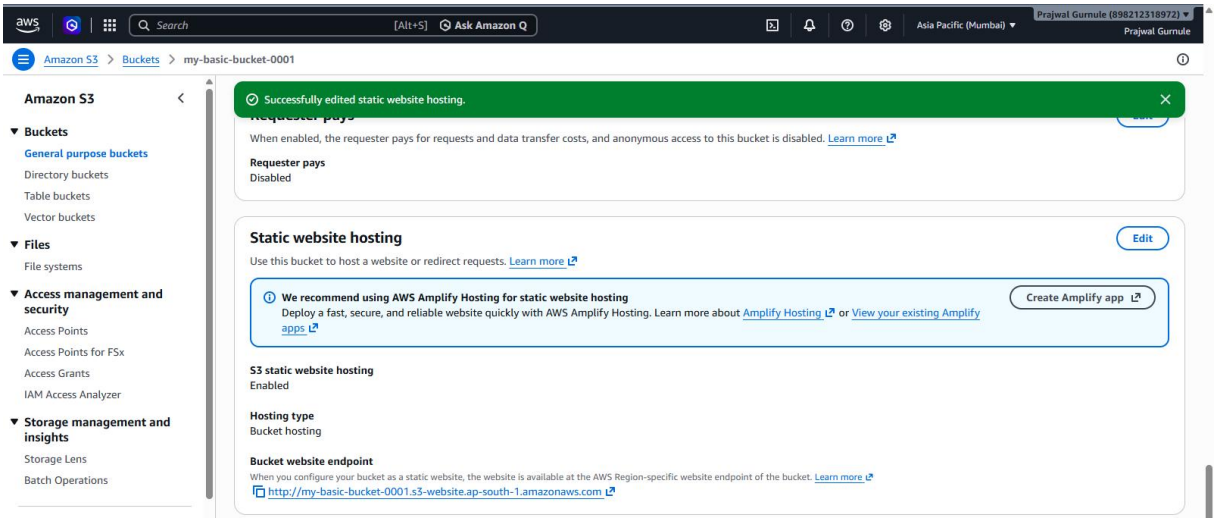
- Go to **Amazon S3**
- Create bucket:
  - Name: `my-basic-bucket-001` (must be unique)
- Open the bucket
- Upload file (example: index.html)
- Click file → copy Object URL
- Make object public (if needed)
- Access file in browser

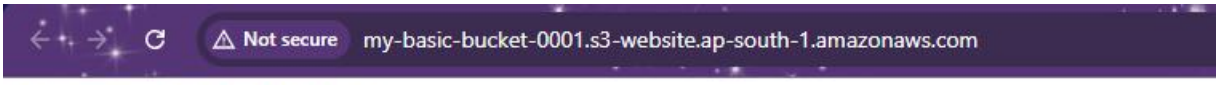












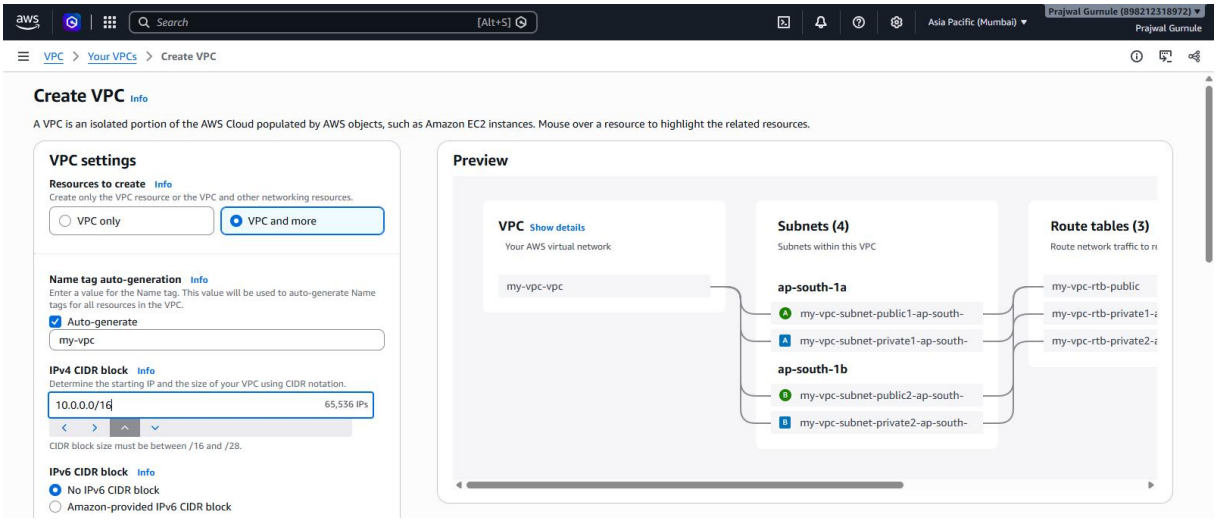
Welcome to our S3 Website!

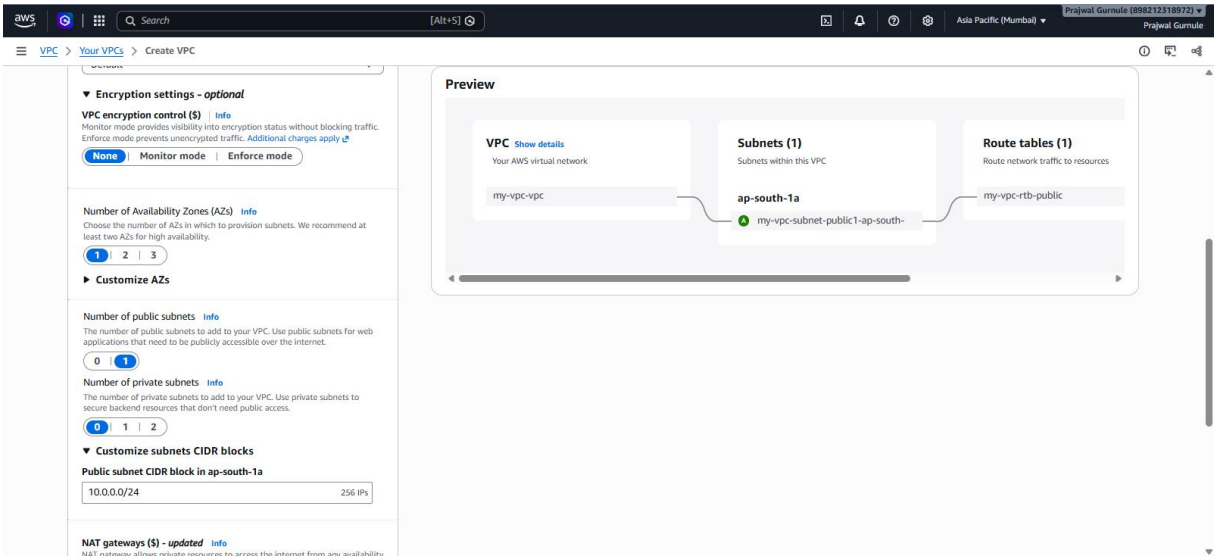
Q3. Create VPC with Public Subnet

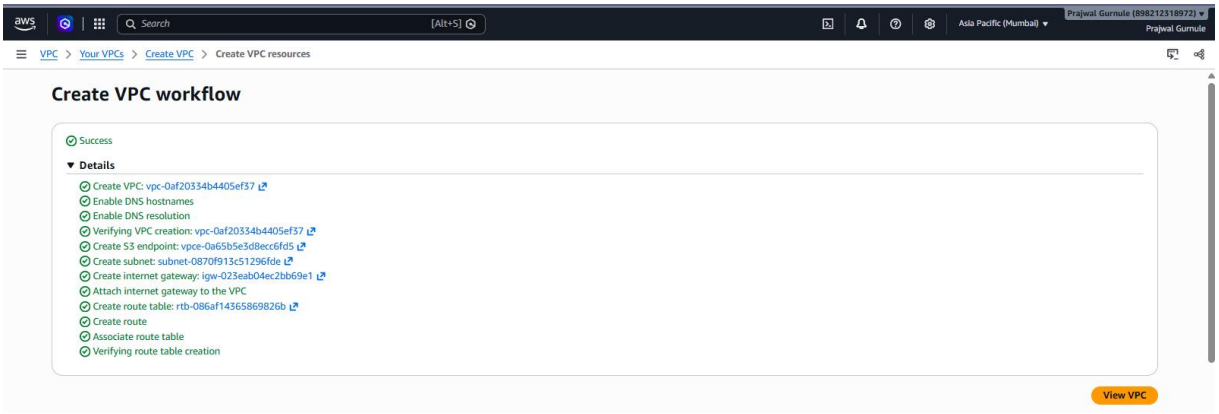
Build your own network.

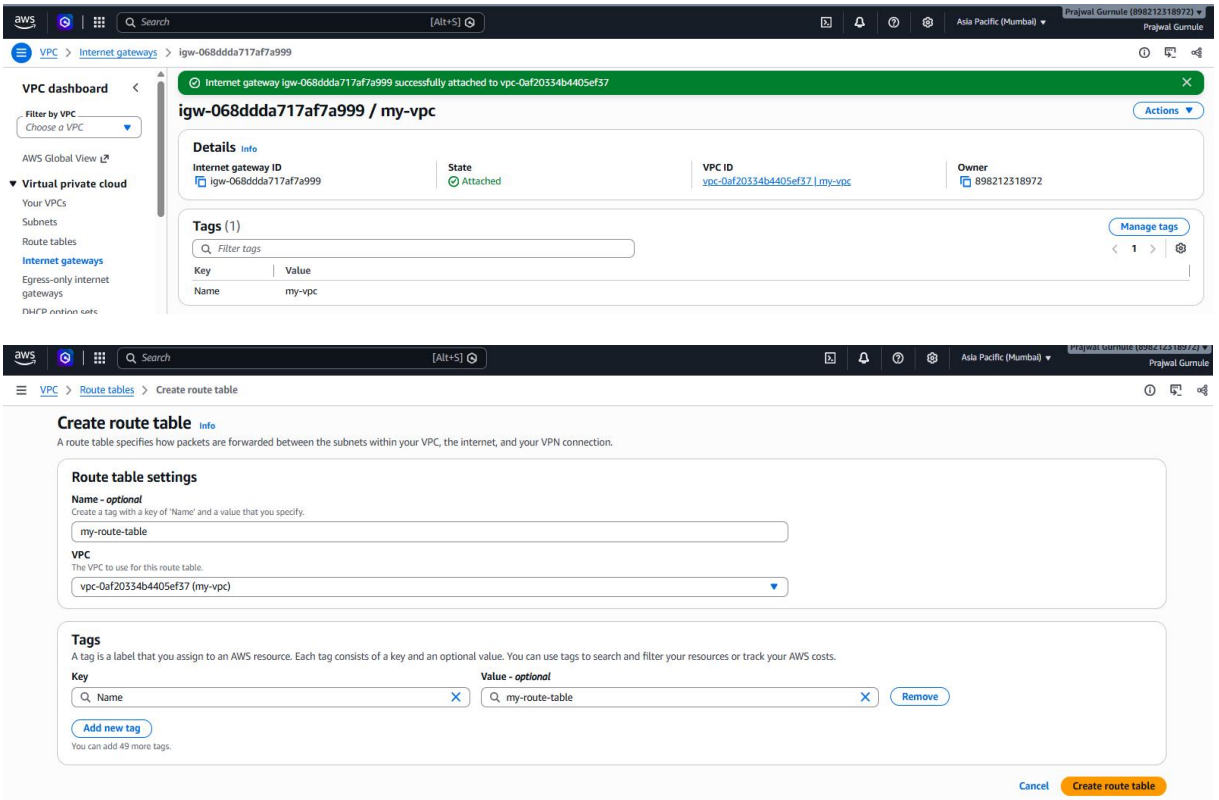
Tasks:

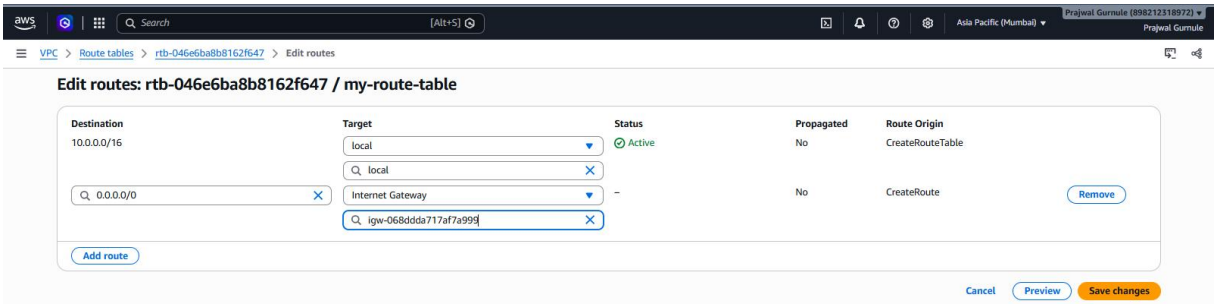
- Go to **Amazon VPC** → Create VPC
  - Name: `my-vpc`
  - CIDR: `10.0.0.0/16`
- Create Subnet:
  - Name: `public-subnet`
  - CIDR: `10.0.1.0/24`
- Create Internet Gateway:
  - Name: `my-igw`
  - Attach to `my-vpc`
- Create Route Table:
  - Add route: `0.0.0.0/0 → Internet Gateway`
  - Associate with public subnet

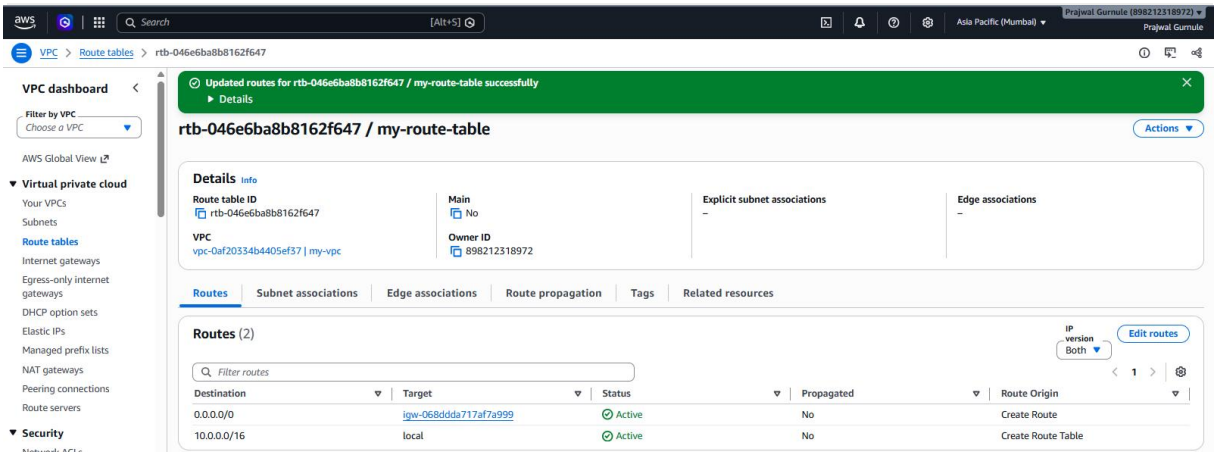



Search
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Asia Pacific (Mumbai)
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VPC > Your VPCs > Create VPC
Encryption settings - optional
VPC encryption control ($) | Info
Monitor mode provides visibility into encryption status without blocking traffic. Enforce mode prevents unencrypted traffic. Additional charges apply
None | Monitor mode | Enforce mode
Number of Availability Zones (AZs) Info
Choose the number of AZs in which to provision subnets. We recommend at least two AZs for high availability.
1 | 2 | 3
Customize AZs
Number of public subnets Info
The number of public subnets to add to your VPC. Use public subnets for web applications that need to be publicly accessible over the internet.
0 | 1
Number of private subnets Info
The number of private subnets to add to your VPC. Use private subnets to secure backend resources that don't need public access.
0 | 1 | 2
Customize subnets CIDR blocks
Public subnet CIDR block in ap-south-1a
10.0.0.0/24
256 IPs
NAT gateways ($) - updated Info
Preview
VPC Show details
Your AWS virtual network
my-vpc-vpc
Subnets (1)
Subnets within this VPC
ap-south-1a
my-vpc-subnet-public1-ap-south-
Route tables (1)
Route network traffic to resources
my-vpc-rtb-public


Search
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Asia Pacific (Mumbai)
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VPC > Your VPCs > Create VPC > Create VPC resources
Create VPC workflow
Success
Details
Create VPC: vpc-0af20334b4405ef37
Enable DNS hostnames
Enable DNS resolution
Verifying VPC creation: vpc-0af20334b4405ef37
Create S3 endpoint: vpce-0a65b5e3d8ecc6fd5
Create subnet: subnet-0870f913c51296fde
Create internet gateway: igw-023eab04ec2bb69e1
Attach internet gateway to the VPC
Create route table: rtb-086af14365869826b
Create route
Associate route table
Verifying route table creation
View VPC


Search
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Asia Pacific (Mumbai)
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VPC > Internet gateways > igw-068ddda717af7a999
VPC dashboard
Filter by VPC
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Internet gateway igw-068ddda717af7a999 successfully attached to vpc-0af20334b4405ef37
igw-068ddda717af7a999 / my-vpc
Actions
Details Info
Internet gateway ID
igw-068ddda717af7a999
State
Attached
VPC ID
vpc-0af20334b4405ef37 | my-vpc
Owner
898212318972
Tags (1)
Manage tags
Filter tags
Key
Value
Name
my-vpc
Search
[Alt+S]
Asia Pacific (Mumbai)
Prajwal Gurnule
VPC > Route tables > Create route table
Create route table Info
A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.
Route table settings
Name - optional
Create a tag with a key of 'Name' and a value that you specify.
my-route-table
VPC
The VPC to use for this route table.
vpc-0af20334b4405ef37 (my-vpc)
Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
Key
Name
Value - optional
my-route-table
Remove
Add new tag
You can add 49 more tags.
Cancel
Create route table





Q4. Launch EC2 Inside Your VPC

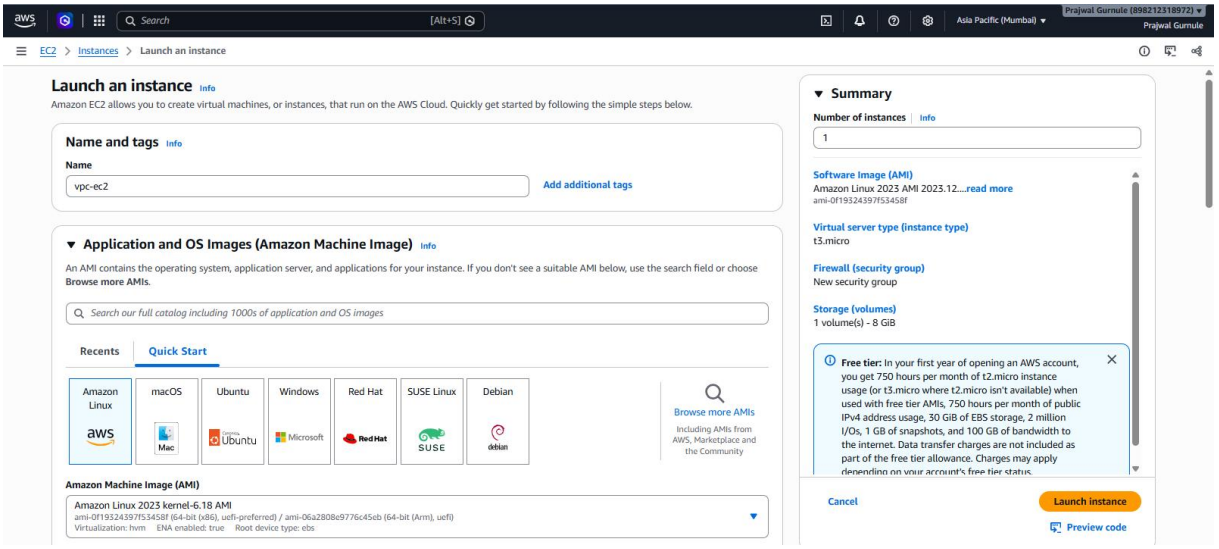
Use your custom network.

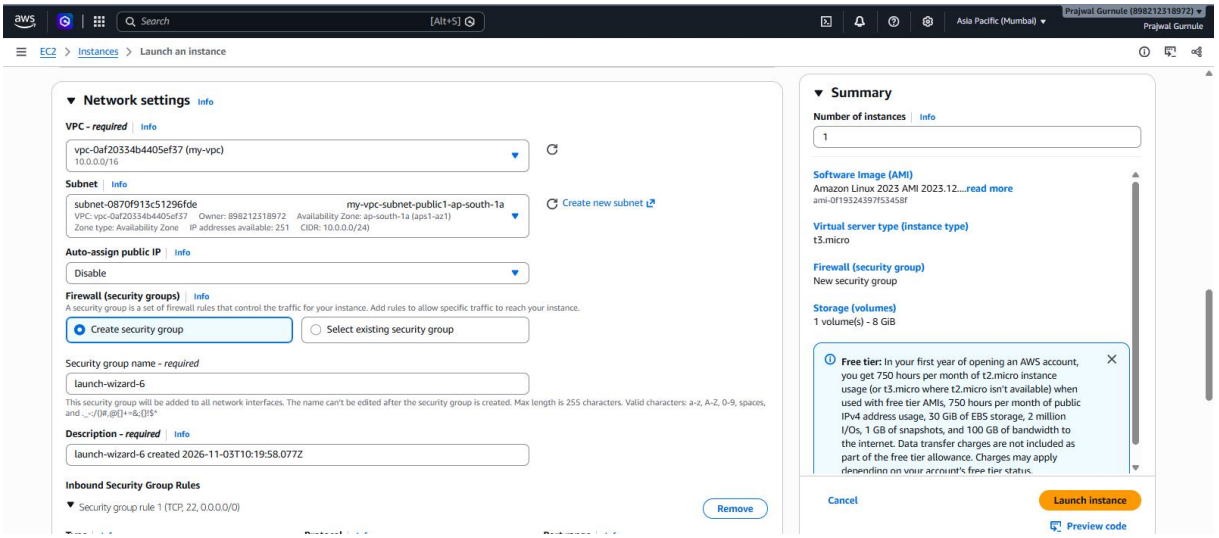
Tasks:

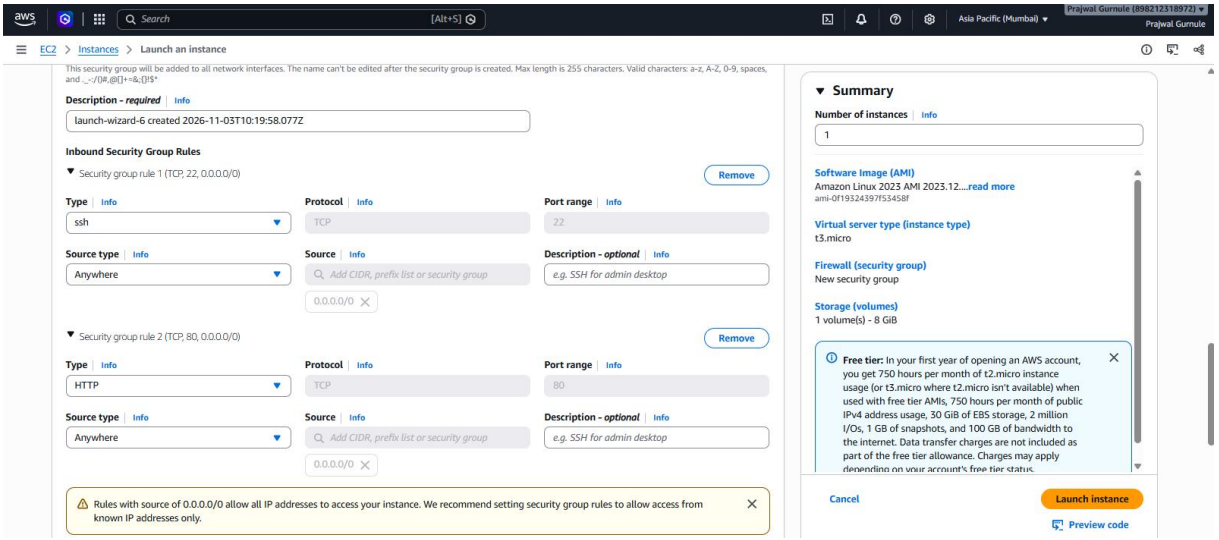
- Go to EC2 → Launch Instance
- Name: `vpc-ec2`
- Select VPC: `my-vpc`
- Select Subnet: `public-subnet`
- Enable Auto-assign Public IP

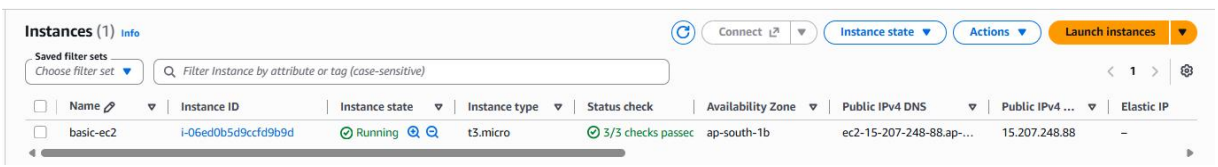
Security Group:

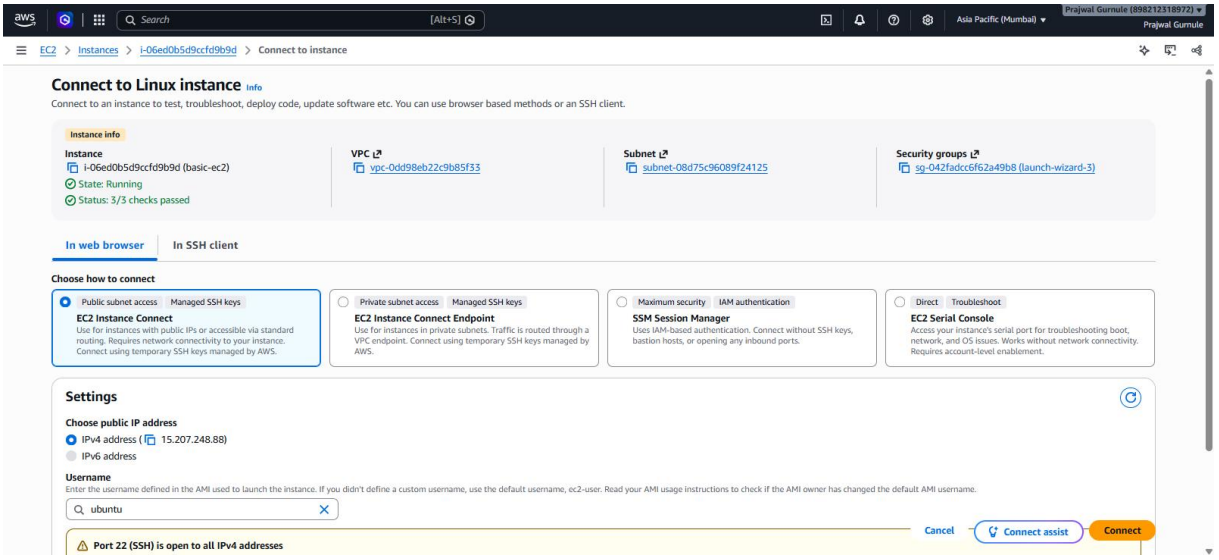
- Allow SSH (22) from your IP
- Allow HTTP (80) from anywhere (`0.0.0.0/0`)
- Launch instance
- Connect using EC2 Instance Connect

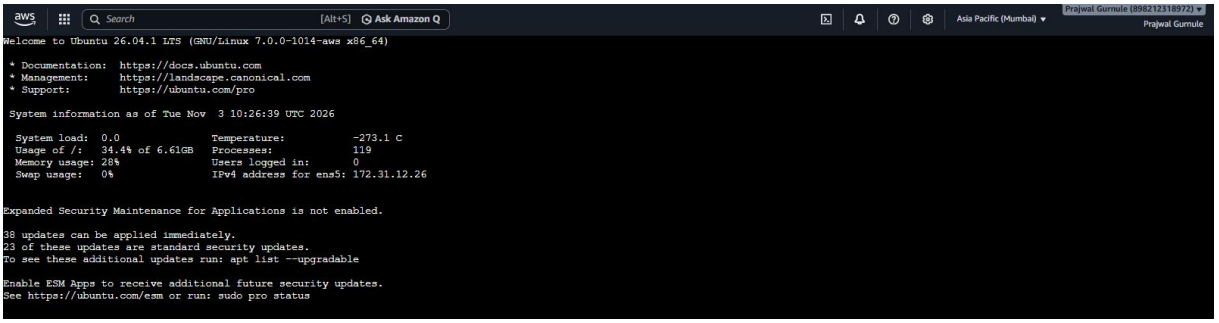

Launch an instance Info
Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.
Name and tags Info
Name
vpc-ec2
Add additional tags
Application and OS Images (Amazon Machine Image) Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Recents
Quick Start
Amazon Linux
macOS
Ubuntu
Windows
Red Hat
SUSE Linux
Debian
Browse more AMIs
Including AMIs from AWS, Marketplace and the Community
Amazon Machine Image (AMI)
Amazon Linux 2023 kernel-6.18 AMI
ami-0f193243971534588f (64-bit (x86), uefi-preferred) / ami-06a2808e9776c45eb (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....read more
ami-0f193243971534588f
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance. Charges may apply depending on your account's free tier status.
Cancel
Launch instance
Preview code


Network settings Info
VPC - required Info
vpc-0af20334b4405ef37 (my-vpc)
10.0.0.0/16
Subnet Info
subnet-0870f913c51296fde my-vpc-subnet-public1-ap-south-1a
Create new subnet
Auto-assign public IP Info
Disable
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
Security group name - required
launch-wizard-6
This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and ._-:/()#,@[]+=&;{}!$*
Description - required Info
launch-wizard-6 created 2026-11-03T10:19:58.077Z
Inbound Security Group Rules
Security group rule 1 (TCP, 22, 0.0.0.0/0)
Remove
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....read more
ami-0f193243971534588f
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance. Charges may apply depending on your account's free tier status.
Cancel
Launch instance
Preview code


This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and ._-:/()#,@[]+=&;{}!$*
Description - required Info
launch-wizard-6 created 2026-11-03T10:19:58.077Z
Inbound Security Group Rules
Security group rule 1 (TCP, 22, 0.0.0.0/0)
Remove
Type Info
ssh
Protocol Info
TCP
Port range Info
22
Source type Info
Anywhere
Source Info
Add CIDR, prefix list or security group
0.0.0.0/0
Description - optional Info
e.g. SSH for admin desktop
Security group rule 2 (TCP, 80, 0.0.0.0/0)
Remove
Type Info
HTTP
Protocol Info
TCP
Port range Info
80
Source type Info
Anywhere
Source Info
Add CIDR, prefix list or security group
0.0.0.0/0
Description - optional Info
e.g. SSH for admin desktop
Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....read more
ami-0f193243971534588f
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance. Charges may apply depending on your account's free tier status.
Cancel
Launch instance
Preview code


Instances (1) Info
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter Instance by attribute or tag (case-sensitive)
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
Public IPv4 DNS
Public IPv4 ...
Elastic IP
basic-ec2
i-06ed0b5d9ccfd9b9d
Running
t3.micro
3/3 checks passed
ap-south-1b
ec2-15-207-248-88.ap-...
15.207.248.88
–





Q5. Attach IAM Role to EC2 and Access S3

Connect EC2 with S3 using role.

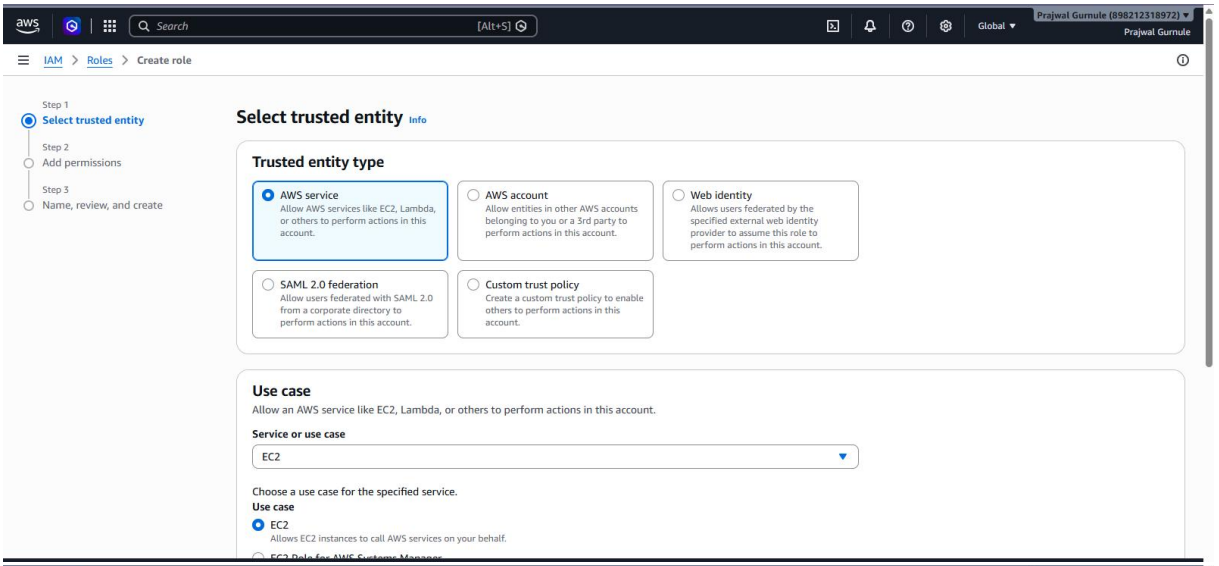
Tasks:

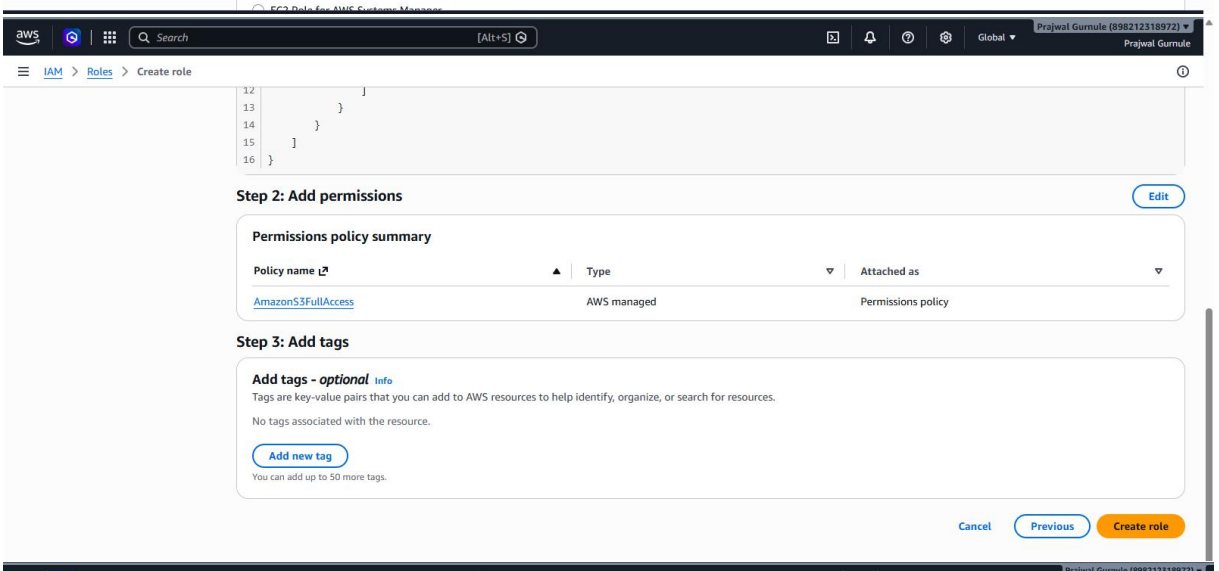
- Go to **AWS IAM** → Create Role
  - Use case: EC2
  - Attach policy: `AmazonS3FullAccess`
- Go to EC2 → Select instance
- Actions → Security → Modify IAM Role
- Attach role

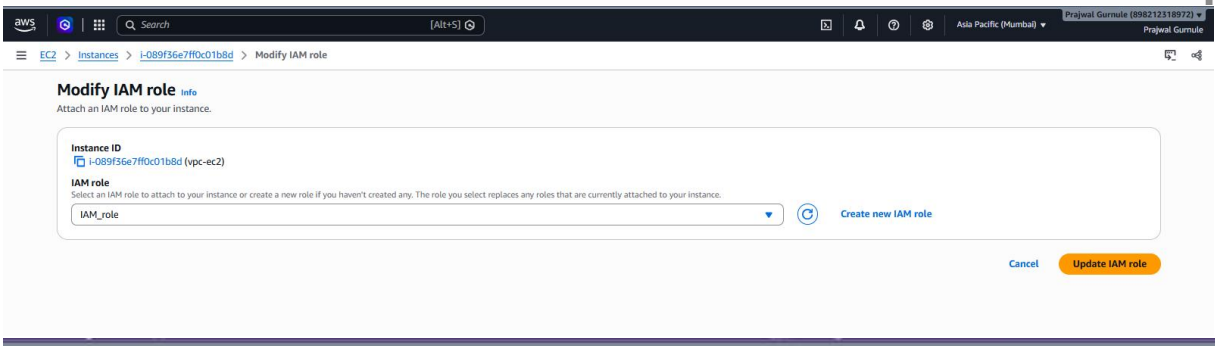
Verification:

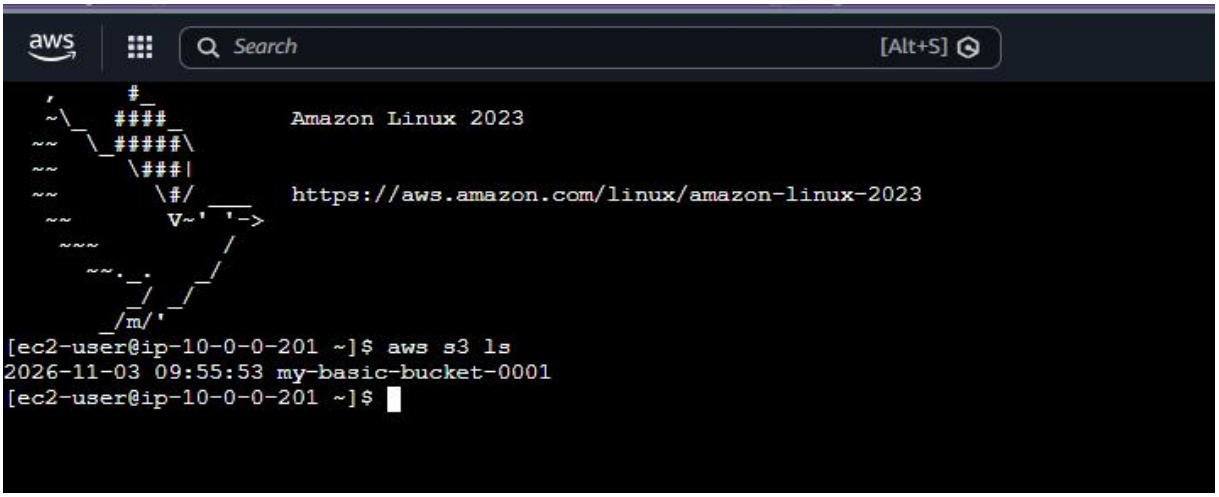
- Connect to EC2 terminal
- Run:

```
aws s3 ls
```


Search [Alt+S]
Global
Prajwal Gurnule (898212318972)
Prajwal Gurnule
IAM > Roles > Create role
Step 1
Select trusted entity
Step 2
Add permissions
Step 3
Name, review, and create
Select trusted entity Info
Trusted entity type
AWS service
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
AWS account
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
Web identity
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
SAML 2.0 federation
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
Custom trust policy
Create a custom trust policy to enable others to perform actions in this account.
Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.
Service or use case
EC2
Choose a use case for the specified service.
Use case
EC2
Allows EC2 instances to call AWS services on your behalf.


Search [Alt+S]
Global
Prajwal Gurnule (898212318972)
Prajwal Gurnule
IAM > Roles > Create role
12]
13 }
14 }
15]
16 }
Step 2: Add permissions
Edit
Permissions policy summary
Policy name
Type
Attached as
AmazonS3FullAccess
AWS managed
Permissions policy
Step 3: Add tags
Add tags - optional Info
Tags are key-value pairs that you can add to AWS resources to help identify, organize, or search for resources.
No tags associated with the resource.
Add new tag
You can add up to 50 more tags.
Cancel
Previous
Create role


Search [Alt+S]
Asia Pacific (Mumbai)
Prajwal Gurnule (898212318972)
Prajwal Gurnule
EC2 > Instances > i-089f36e7ff0c01b8d > Modify IAM role
Modify IAM role Info
Attach an IAM role to your instance.
Instance ID
i-089f36e7ff0c01b8d (vpc-ec2)
IAM role
Select an IAM role to attach to your instance or create a new role if you haven't created any. The role you select replaces any roles that are currently attached to your instance.
IAM_role
Create new IAM role
Cancel
Update IAM role


Search [Alt+S]
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-10-0-0-201 ~]$ aws s3 ls
2026-11-03 09:55:53 my-basic-bucket-0001
[ec2-user@ip-10-0-0-201 ~]$

Q6. Create Private Subnet + EC2 (No Internet)

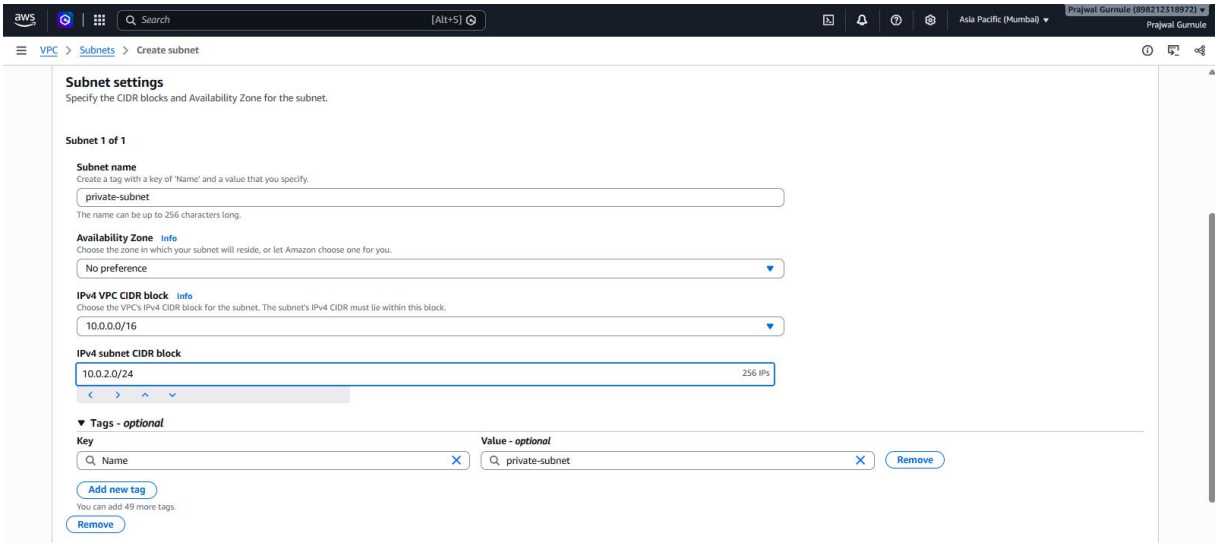
Secure backend server.

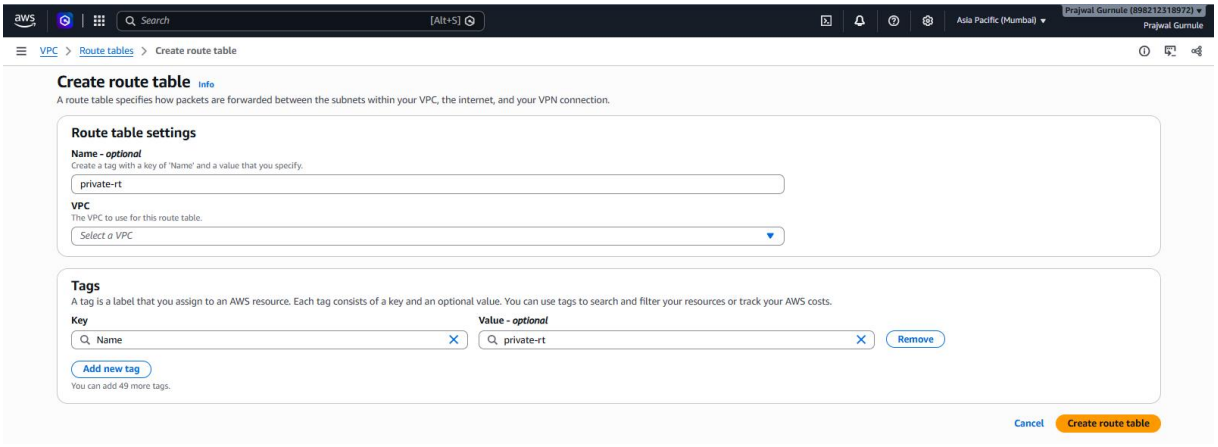
Tasks:

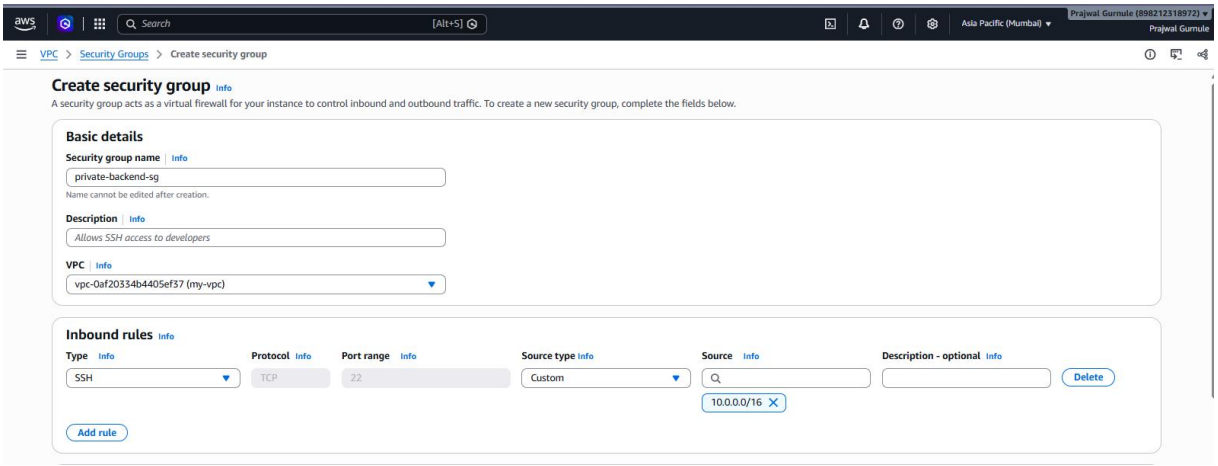
- Create Subnet:
  - Name: `private-subnet`
  - CIDR: `10.0.2.0/24`
- Create Route Table:
  - Name: `private-rt`
  - Do not add Internet Gateway route
- Associate with private subnet
- Launch EC2:
  - Name: `private-ec2`
  - Subnet: `private-subnet`
  - Disable public IP

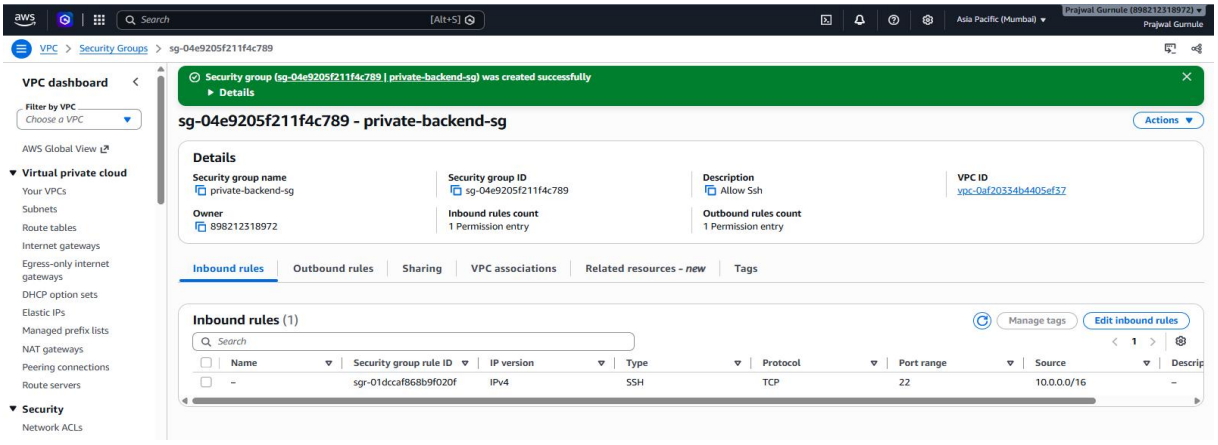
Security Group:

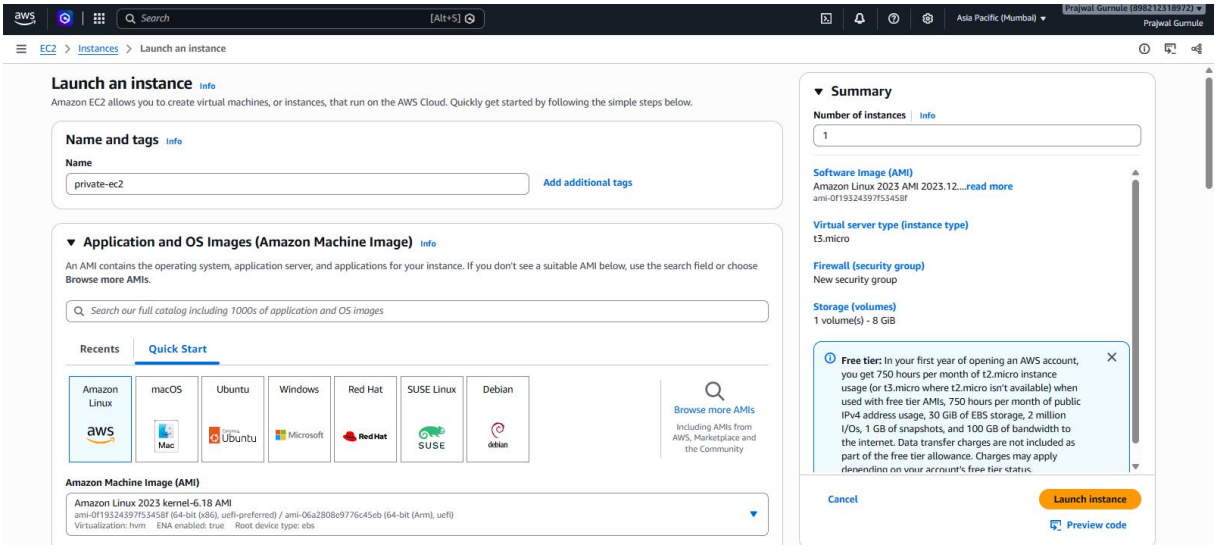
- Allow SSH only from VPC CIDR (`10.0.0.0/16`)

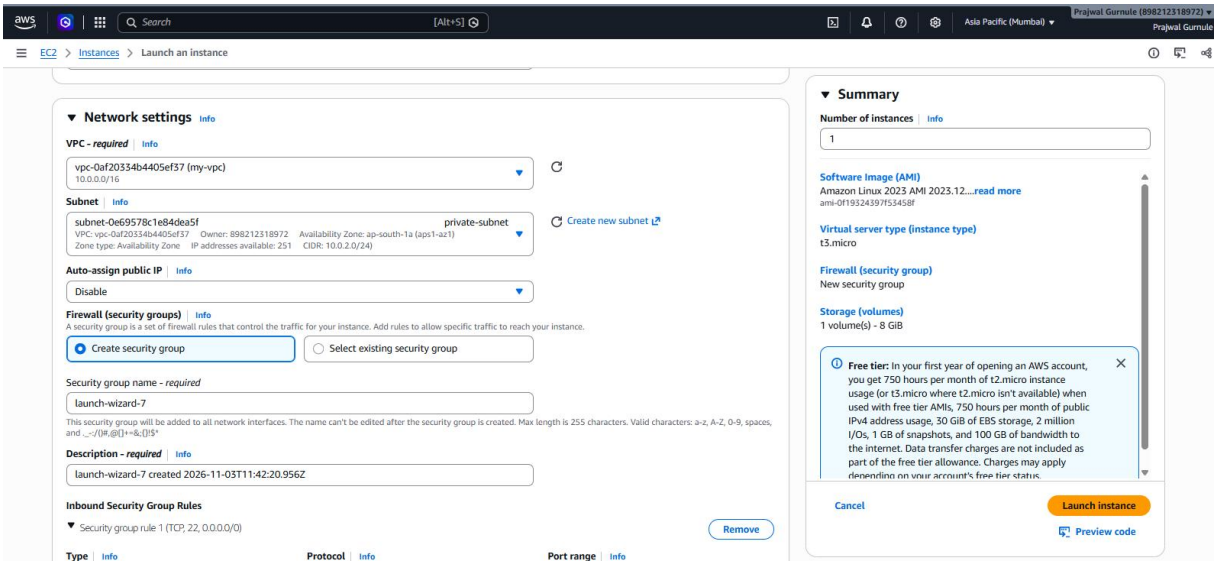


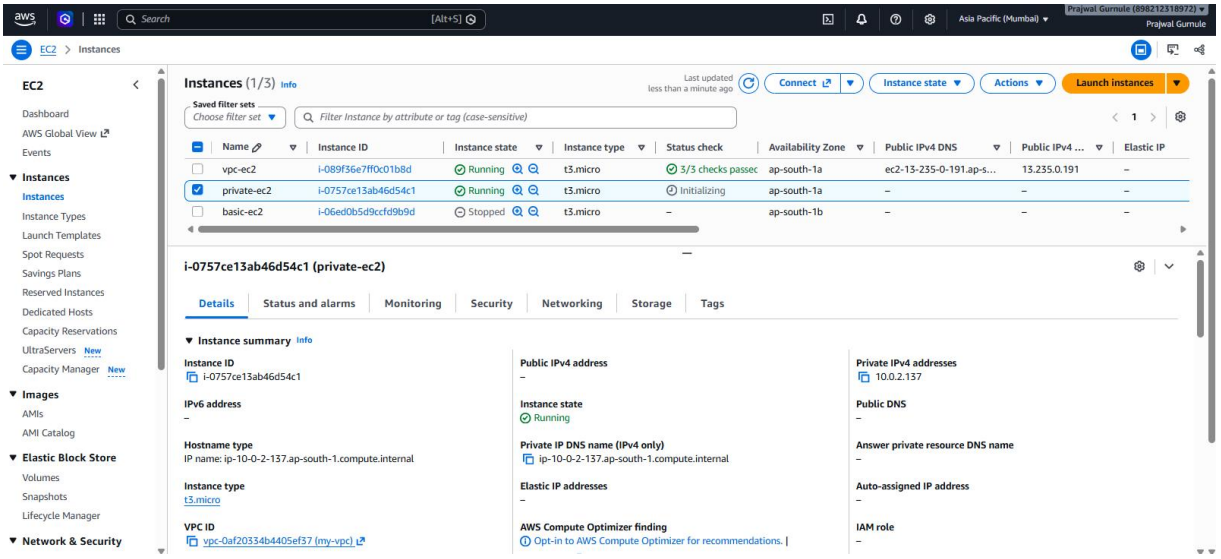

VPC > Route tables > Create route table
Create route table Info
A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.
Route table settings
Name - optional
Create a tag with a key of 'Name' and a value that you specify.
private-rt
VPC
The VPC to use for this route table.
Select a VPC
Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
Key
Name
Value - optional
private-rt
Remove
Add new tag
You can add 49 more tags.
Cancel
Create route table


VPC > Security Groups > Create security group
Create security group Info
A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.
Basic details
Security group name Info
private-backend-sg
Name cannot be edited after creation.
Description Info
Allows SSH access to developers
VPC Info
vpc-0af20334b4405ef37 (my-vpc)
Inbound rules Info
Type Info
Protocol Info
Port range Info
Source type Info
Source Info
Description - optional Info
SSH
TCP
22
Custom
10.0.0.0/16
Delete
Add rule


VPC > Security Groups > sg-04e9205f211f4c789
Security group (sg-04e9205f211f4c789 | private-backend-sg) was created successfully
Details
VPC dashboard
sg-04e9205f211f4c789 - private-backend-sg
Actions
Details
Security group name
private-backend-sg
Security group ID
sg-04e9205f211f4c789
Description
Allow Ssh
VPC ID
vpc-0af20334b4405ef37
Owner
898212318972
Inbound rules count
1 Permission entry
Outbound rules count
1 Permission entry
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (1)
Manage tags
Edit inbound rules
Name
Security group rule ID
IP version
Type
Protocol
Port range
Source
sgr-01dccaf868b9f020f
IPv4
SSH
TCP
22
10.0.0.0/16


EC2 > Instances > Launch an instance
Launch an instance Info
Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.
Name and tags Info
Name
private-ec2
Add additional tags
Application and OS Images (Amazon Machine Image) Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Recents
Quick Start
Amazon Linux
macOS
Ubuntu
Windows
Red Hat
SUSE Linux
Debian
Browse more AMIs
Including AMIs from AWS, Marketplace and the Community
Amazon Machine Image (AMI)
Amazon Linux 2023 kernel-6.18 AMI
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....read more
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance. Charges may apply
Cancel
Launch instance
Preview code





Q7. Create Bastion Host (Jump Server)

Access private EC2 securely.

Tasks:

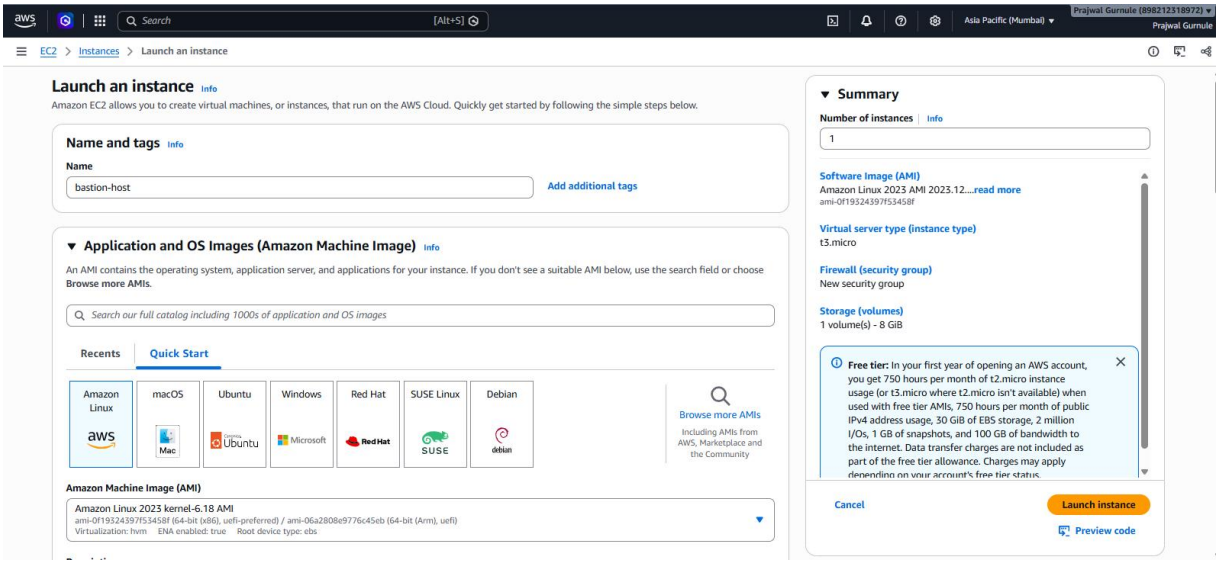
- Launch EC2:
  - Name: `bastion-host`
  - Subnet: `public-subnet`
  - Enable public IP

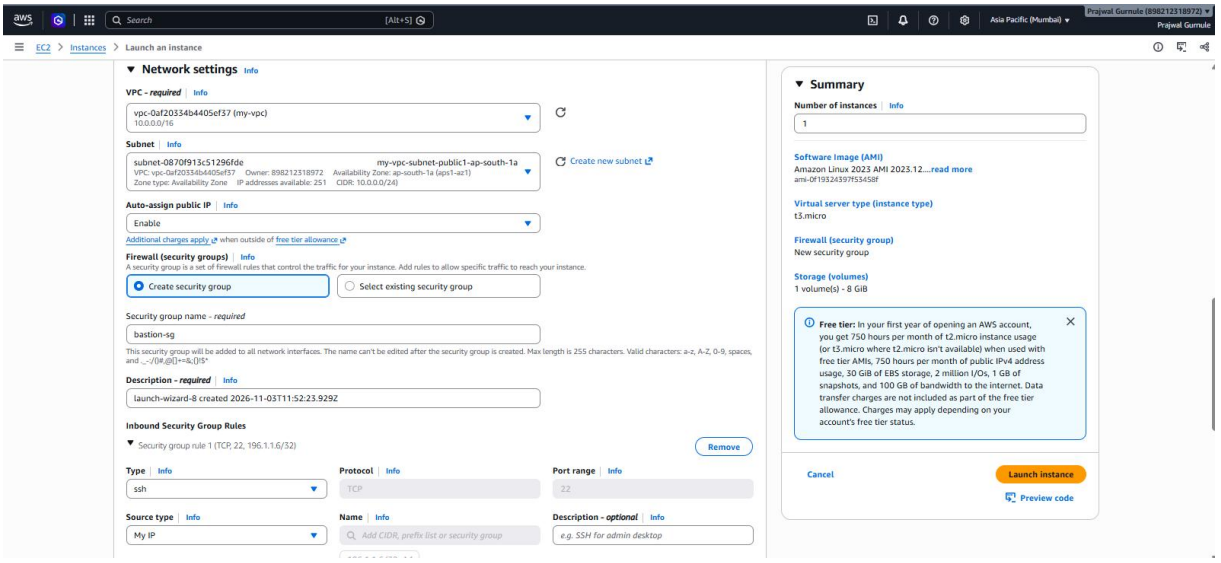
Security Group (Bastion):

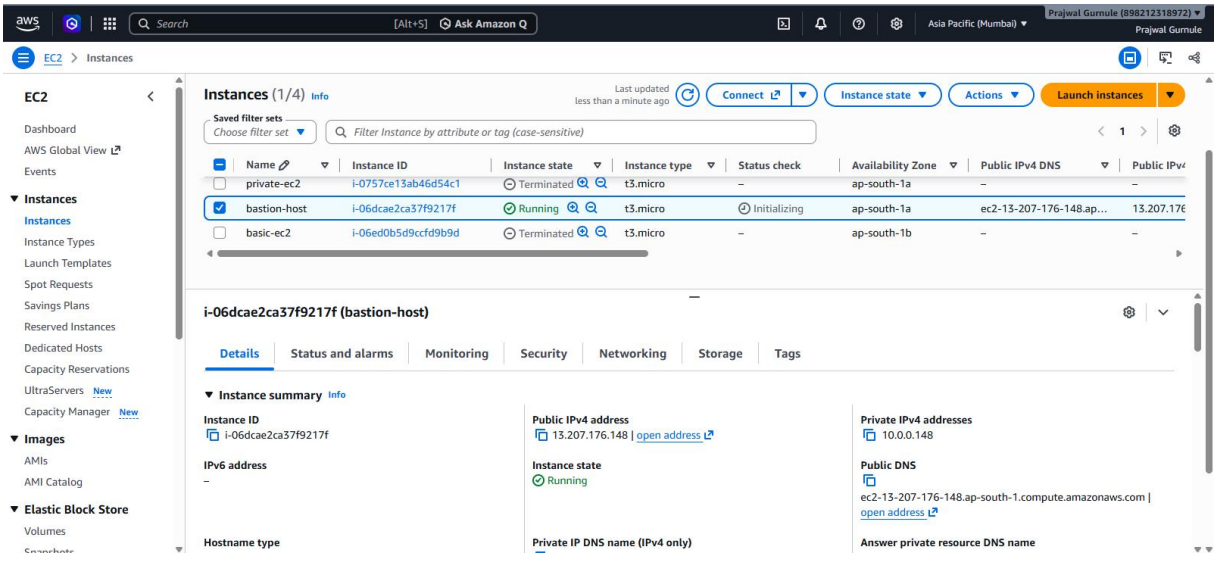
- Allow SSH (22) from your IP

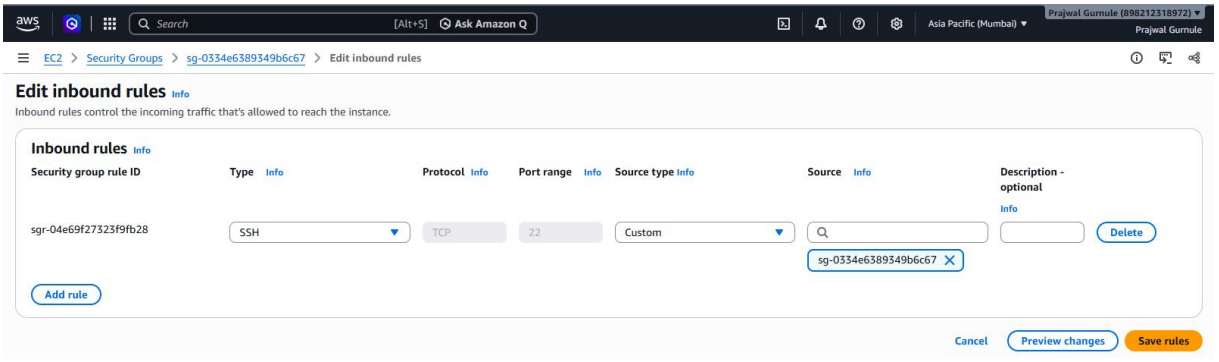
Modify Private EC2 Security Group:

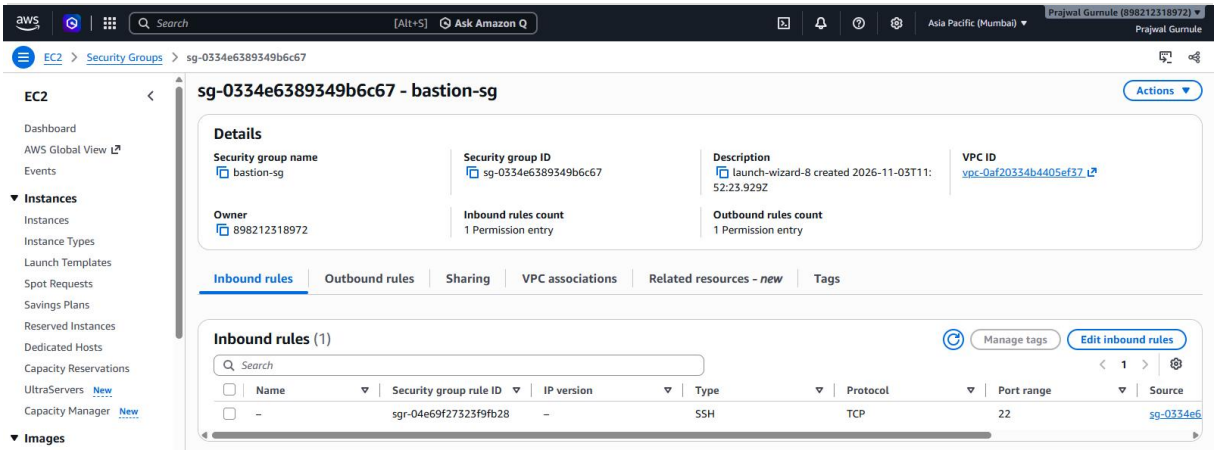
- Allow SSH only from Bastion Security Group (not IP)


Launch an instance
Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.
Name and tags
Name
bastion-host
Add additional tags
Application and OS Images (Amazon Machine Image)
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Recents
Quick Start
Amazon Linux
macOS
Ubuntu
Windows
Red Hat
SUSE Linux
Debian
Browse more AMIs
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Amazon Linux 2023 kernel-6.18 AMI
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t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance. Charges may apply
Cancel
Launch instance
Preview code


Network settings
VPC - required
vpc-0af20334b4405ef37 (my-vpc)
10.0.0.0/16
Subnet
subnet-0870f913c51296fde
my-vpc-subnet-public1-ap-south-1a
Create new subnet
Auto-assign public IP
Enable
Firewall (security groups)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
Security group name - required
bastion-sg
Description - required
launch-wizard-8 created 2026-11-03T11:52:23.929Z
Inbound Security Group Rules
Security group rule 1 (TCP, 22, 196.1.1.6/32)
Remove
Type
ssh
Protocol
TCP
Port range
22
Source type
My IP
Name
Add CIDR, prefix list or security group
Description - optional
e.g. SSH for admin desktop
Summary
Number of instances
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....read more
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Cancel
Launch instance
Preview code


EC2
Instances
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Instances (1/4)
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter Instance by attribute or tag (case-sensitive)
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
Public IPv4 DNS
private-ec2
i-0757ce13ab46d54c1
Terminated
t3.micro
ap-south-1a
bastion-host
i-06dcae2ca37f9217f
Running
t3.micro
Initializing
ap-south-1a
ec2-13-207-176-148.ap...
basic-ec2
i-06ed0b5d9ccfd9b9d
Terminated
t3.micro
ap-south-1b
i-06dcae2ca37f9217f (bastion-host)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary
Instance ID
i-06dcae2ca37f9217f
Public IPv4 address
13.207.176.148 | open address
Private IPv4 addresses
10.0.0.148
IPv6 address
Instance state
Running
Public DNS
ec2-13-207-176-148.ap-south-1.compute.amazonaws.com | open address
Hostname type
Private IP DNS name (IPv4 only)
Answer private resource DNS name





Q8. Verify Private Access via Bastion

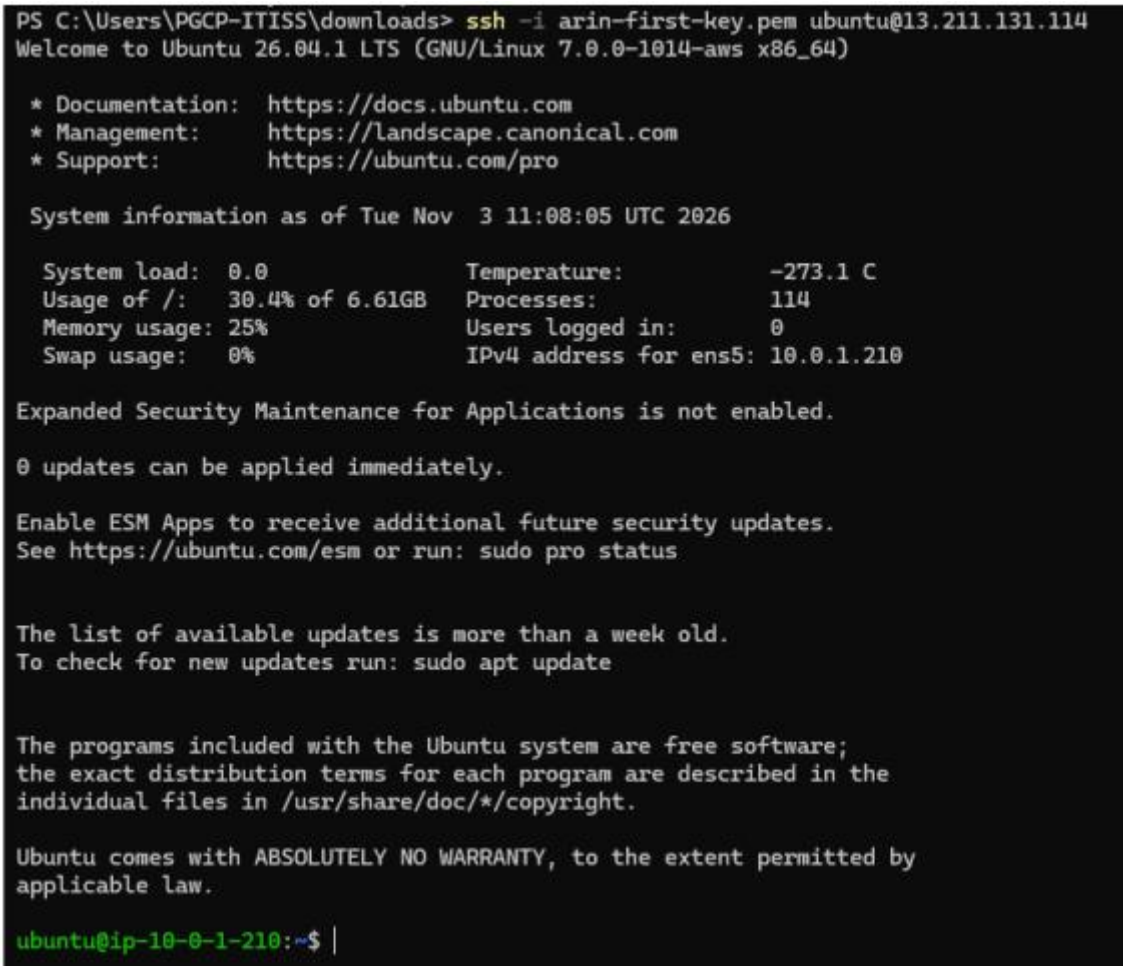
Tasks:

- Connect to Bastion host
- From Bastion:

```
ssh -i key.pem ubuntu@<private-ec2-private-ip>
```

Observation:

- Private EC2 has no public IP
- Access only through Bastion

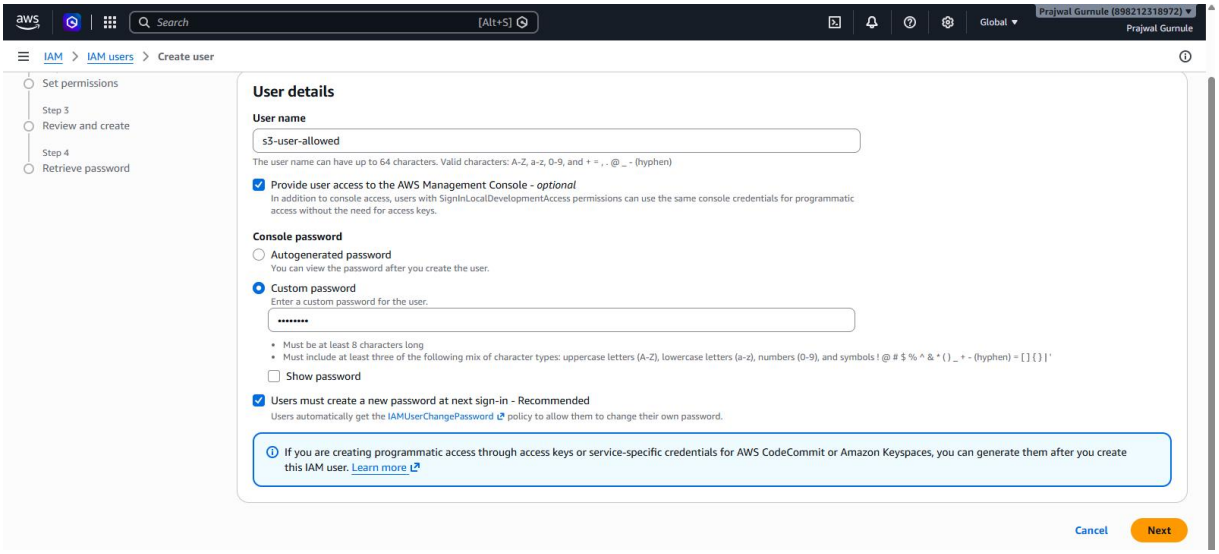

PS C:\Users\PGCP-ITISS\downloads> ssh -i arin-first-key.pem ubuntu@13.211.131.114
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-1014-aws x86_64)
* Documentation: https://docs.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
System information as of Tue Nov 3 11:08:05 UTC 2026
System load: 0.0
Usage of /: 30.4% of 6.61GB
Memory usage: 25%
Swap usage: 0%
Temperature: -273.1 C
Processes: 114
Users logged in: 0
IPv4 address for ens5: 10.0.1.210
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
The list of available updates is more than a week old.
To check for new updates run: sudo apt update
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
ubuntu@ip-10-0-1-210:~$

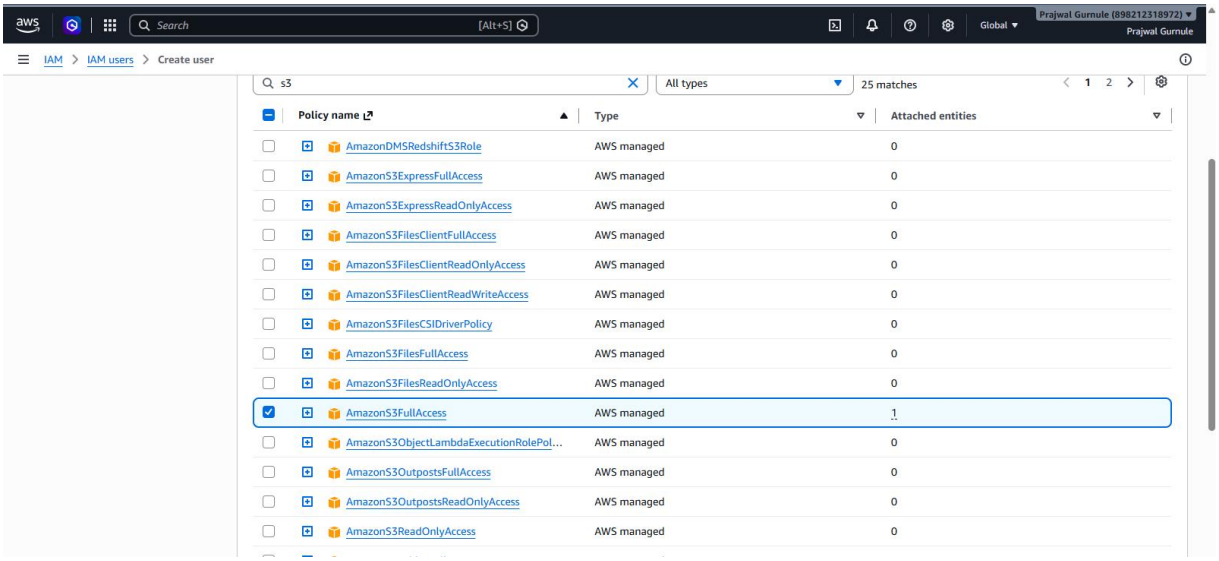
Q9. Create Two IAM Users (Access vs No Access)

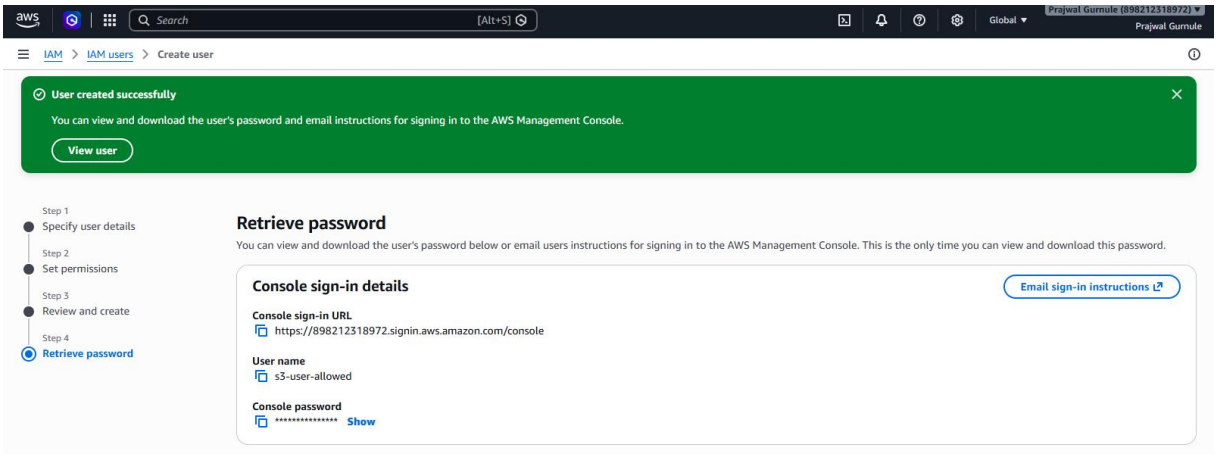
Test permission control.

Tasks:

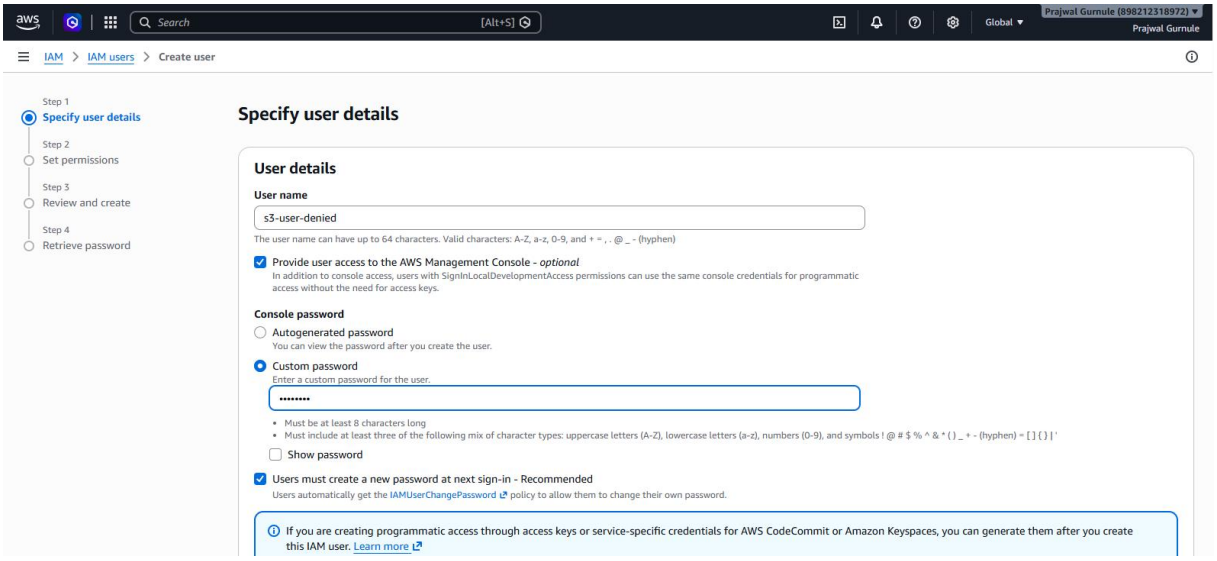
- Create User1:
  - Name: `s3-user-allowed`
  - Enable console access
  - Attach policy: `AmazonS3FullAccess`

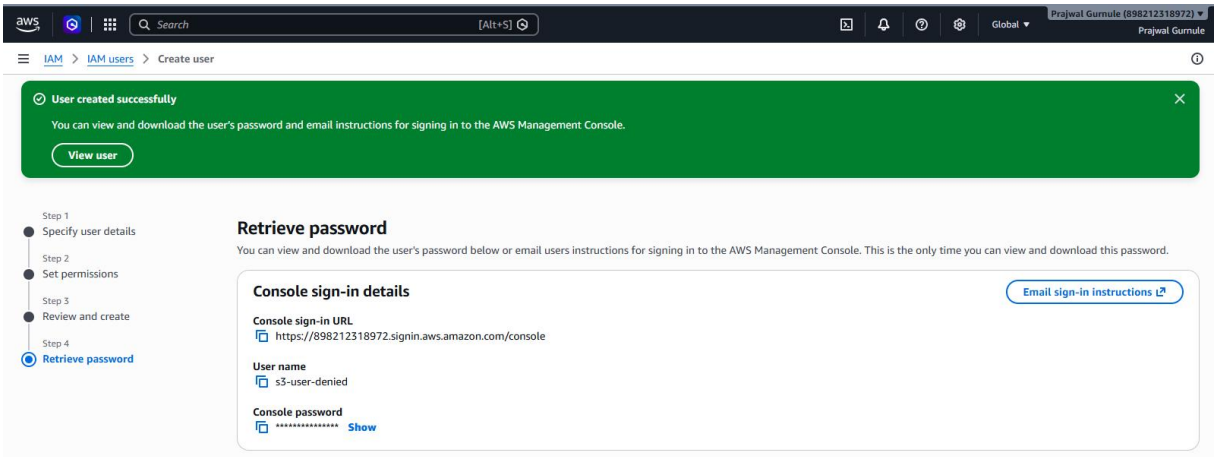






- Create User2:
  - Name: `s3-user-denied`
  - Enable console access
  - Do not attach any policy



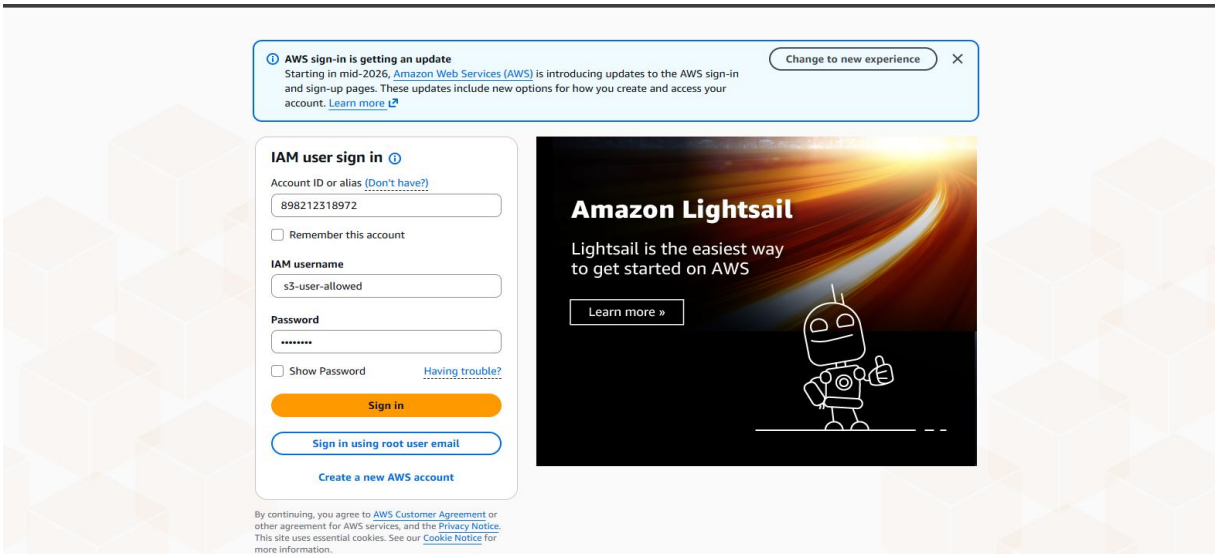


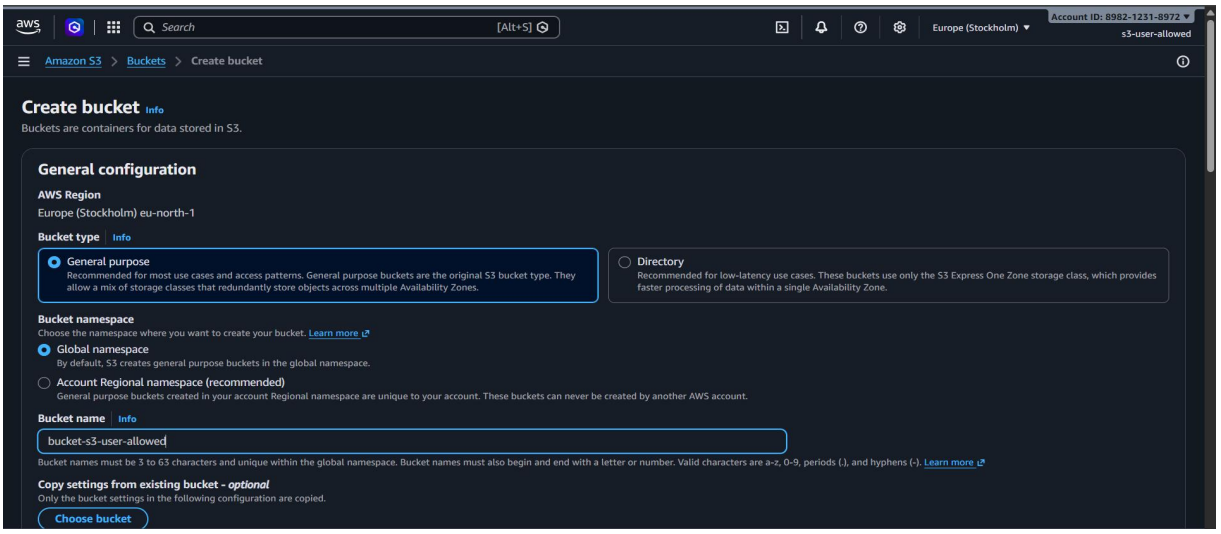
Q10. Test S3 Access (Permission Check)

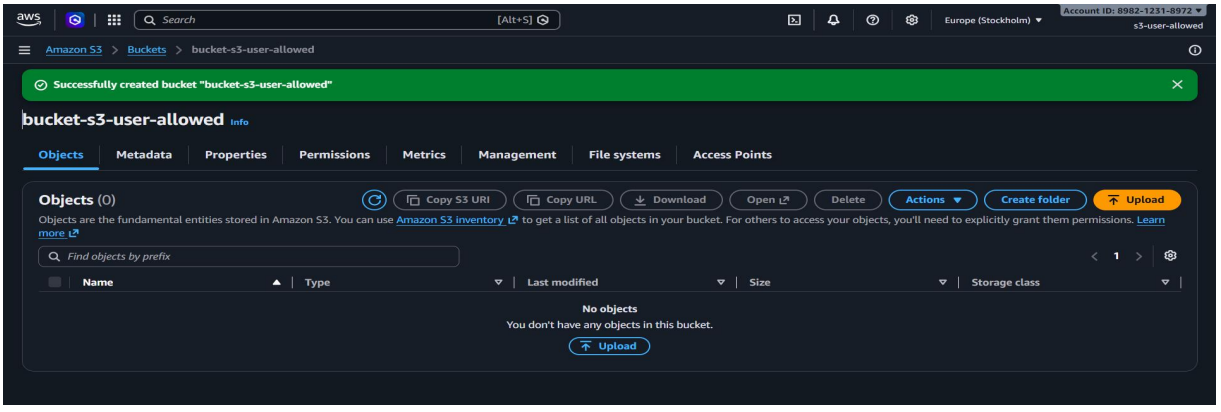
Tasks:

Login as User1:

- Go to S3
- Upload file
- Should succeed







Login as User2:

- Go to S3
- Try to upload file
- Should fail (Access Denied)

